

Inversion sequences avoiding 100, 101, 201 (OEIS A279571)

The leading equivalent and the cone exponent

Two research reports of one external research session
(bundle Reports 125 and 123), merged into one ProveIt report

Parts I and II: 2 October 2026 merged: 5 October 2026

Abstract

Let a_n count the inversion sequences of length n that avoid the patterns 100, 101 and 201: OEIS A279571, Class 2106 of Britt and Beaton. A positive change of weights turns Britt and Beaton's generating tree into a centred Markov-additive walk with two colours and unbounded steps in the quadrant; its effective covariance gives the wedge angle $\theta = \arccos(2/\sqrt{7})$ and the irrational exponent $p = \pi/\theta \in (4, 5)$. This report prints two manuscripts as Parts I and II. Part I (Report 125) proves $a_n \sim C_A 9^n n^{-\kappa}$ with $\kappa = 1 + p \approx 5.40169$ and $0 < C_A < \infty$, where C_A is characterized by normalized forward and reversed killed harmonic functions and a positive endpoint series; on the way it proves a killed local limit theorem at every fixed pair of states. It also proves a threshold inverse with an explicit real centre and a ceiling bracket of vanishing width, that $1/9$ is the only singularity of the generating function on its circle of convergence, and that the generating function is not D-finite over $\mathbb{C}(z)$. Part II (Report 123, whose archive was created 49 minutes earlier) proves the weaker law $a_n = 9^n n^{-\kappa+o(1)}$ by geometric annuli and exact-time gluing; it is the earlier proof of the exponent, of the circle statement and of non-D-finiteness, and it alone has the functional equations, kernel and discriminant of the generating tree. A remark added in the merge shows that Report 123's walk satisfies hypotheses (H1)–(H4) of the logarithmic cone theorem of the neighbouring report [a377922-corner-polyhedra-schnyder](#) (its Theorem 24.1) but not the seed hypothesis (H5) at the origin endpoints that the theorem states, since no legal path returns to the origin; the theorem's proof, run at fixed endpoint states, gives Report 123's exponent. No amplitude digits, remainder rate, correction expansion or local expansion at $1/9$ is proved. The report is unrefereed, and nothing in it is formalized.

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Guide to this report

The two Parts are two manuscripts of the session bundle of Reports 1–243 (arrival commit 60f54ea06), placed in this directory by commit 612787fb4 (batch 104). Both are dated 2 October 2026. Report 123 is the earlier (its archive was created at 14:23:20 UTC, Report 125’s at 15:12:25 UTC), but Report 125 proves every statement of Report 123’s main theorem or a stronger one, under the same hypotheses, so it is the base and comes first:

Part	Bundle report	Delivered title	Labels
I	125 (base)	A279571: the leading equivalent. A positive harmonic constant and a sharp threshold inverse	<code>icw:</code>
II	123	A279571 exact logarithmic exponent. Unique dominant singularity and non D finiteness	<code>icw:log:</code>

Each Part prints its manuscript’s text, with every result, proof, remark and limitation; statements keep their delivered wording. Sections and statements are numbered continuously, statements within their sections; both deliveries number equations consecutively through the document, and so does this report. Part I keeps Report 125’s section, statement and equation numbers (Sections 1–14); Section 14.1 is added in the write. In Part II, Report 123’s Section k is Section $k + 14$ and its Lemma $k.j$ is Lemma $(k + 14).j$; its equations continue the report’s numbering, so their numbers differ from the delivery (the README maps them). About two fifths of Report 123 is word for word Report 125’s text: its Sections 2.1, 3.1, 3.2, 4.2, 8 and 9 are Report 125’s Sections 2.1, 3.1, 3.2, 9.1, 12 and 13. That text is printed once, in Part I, and Part II keeps the headings with pointers (Sections 16.1, 17.1, 17.2, 18.2, 22 and 23). Part II ends with two sections added in the write, Sections 26 and 27.

Notes added when the manuscripts were merged are set in small type and tagged [**write**], with their date. They give cross-references between the Parts, mark results proved twice, mark statements that Part I supersedes, and relate the report to its neighbours. The write adds one proposition with its proof, Proposition 26.2, a remark with a proof, Remark 26.1, and a corollary, Corollary 26.3, all in Section 26. Superseded statements of Part II are printed as delivered, with a dated note pointing to Part I, as in the merged report `a135501-closed-lambda-terms` of the same collection directory. No delivered statement was changed, and no symbol was renamed. Each Part ends with a “Further questions and research” section (Sections 14.1 and 27), which collects, under Vladimir’s standing rule of 4 October 2026, every claim of that Part that is stated without proof, with its source, its sketch and what is missing.

What is proved, and where

Here $p = \pi / \arccos(2/\sqrt{7})$, $\kappa = 1 + p$, K_n is the killed kernel of the tilted walk, $o = (0, 0, P)$ its start and $e = (1, 0, P)$.

Statement	Where
$a_n \sim C_A 9^n n^{-\kappa}$ with $0 < C_A < \infty$ and C_A given by the series (10)	Part I, Theorem 1.1

Statement	Where
Normalized killed harmonic functions V, V^* with $V = U + O((1 + s)^{p-1+\delta})$; the killed local limit $K_n((s, c), (t, d)) \sim (b_\theta \pi_d / \sqrt{\det \Sigma}) V(s, c) V^*(t, d) n^{-p-1}$ at fixed states; $K_n(o, (t, d)) \leq C(1 + t)^{2p} n^{-p-1}$	Part I, Theorem 1.1, Sections 7–10
Threshold inverse $\lceil v(y) - \varepsilon(y) \rceil \leq N(y) \leq \lceil v(y) + \varepsilon(y) \rceil$, $\varepsilon(y) \rightarrow 0$, explicit centre $v(y)$	Part I, Corollary 1.2; it also follows from Theorem 20.1 of a377922-corner-polyhedra-schnyder
Radius $1/9$; $1/9$ the only singularity on $ z = 1/9$; the log-limit of $F^{(5)}$; p, κ irrational; F not D-finite over $\mathbb{C}(z)$	Part I, Corollary 12.1, Sections 12–13 (Report 123’s Sections 8–9, proved there first, from the logarithmic law)
Weaker: $c_\varepsilon 9^n n^{-\kappa-\varepsilon} \leq a_n \leq C_\varepsilon 9^n n^{-\kappa+\varepsilon}$; $N(y) = \frac{\log y}{\log 9} + \frac{\kappa}{\log 9} \log \log y + o(\log \log y)$	Part II, Theorem 15.1 (second route: functional limit theorem, annuli, exact-time gluing)
Functional equations of the generating tree, the kernel, its discriminant, the branch points and their collision at $(1/9, 3, 5/3)$	Part II, Section 16.2 (not in Part I)
Exact-time interior bridge; uniform annulus estimates; survival from starts at radius $O(\log n)$	Part II, Lemmas 18.1, 19.1, 20.1
[write] (H1)–(H4) of Theorem 24.1 of a377922-corner-polyhedra-schnyder hold for this walk; (H5) fails at the origin endpoints, where $K_n(o, ((0, 0), b)) = 0$ for $n \geq 1$	Remark 26.1
[write] That theorem at fixed endpoint states, from its proof; Report 123’s exponent as a corollary	Proposition 26.2, Corollary 26.3
<i>Open in both Parts</i>	
Digits of C_A or an effective evaluation of $V(o)$ and the endpoint series; any rate for $a_n/(C_A 9^n n^{-\kappa}) - 1$, a $1/n$ or other correction, a correction transseries; a local Δ -domain expansion at $1/9$; an unconditional rule $N(y) = \lceil v(y) \rceil$; effective onsets; a general finite-colour cone theorem with amplitudes. See Sections 14.1 and 27.	

Finite computations in both Parts (enumerations, exact rational algebra, mutation tests) are checks of formulas and of the combinatorial model; no asymptotic statement rests on them. The literature statements that the Parts use (strong approximation, Brownian wedge kernels, the martingale functional limit theorem, the G -function theorem, Britt and Beaton’s generating tree) were not re-checked against the papers at the write, except the locators of Fischler and Rivoal, which the independent check of **a377922-corner-polyhedra-schnyder** confirmed. The generating tree itself is re-derived in Section 2.1; the delivered checkers compare it with direct pattern-avoidance enumeration through $n = 9$ and with the OEIS terms through $n = 65$, and the intake compared the walk with the OEIS terms through $n = 90$.

Notation across the two Parts

Common to both Parts, with the same meaning: $\theta, p, \kappa = 1 + p$, the quadrant $\mathcal{Q} = \mathbb{Z}_{\geq 0}^2$, the colours P, Q , the start $o = (0, 0, P)$, the steps A_d, C_d, D_j , the tilted law q_{cd} , the Perron vector $r = (2, 1)$,

the endpoint weight g , the killed kernels K_m, K_m^* , the colour matrix P_0 , $\pi = (2/3, 1/3)$, the drifts m_c , the correctors h_c, h_c^* , the covariances Γ_c and Σ , the dual law q^* , the Fourier matrix $L(u)$, the killed operator T on $\ell^\infty(\mathcal{D})$ and the row functional v of the circle sections, $\epsilon_l = 9^{-l}$, $\lambda = \log 9$ and $L = \log y$ in the inverse, and $N(y) = \min\{n : a_n \geq y\}$. The letters below change meaning between the Parts or inside one; each Part's own text fixes its meaning there.

Labels. Twenty-five label names occur in both manuscripts. Under the prefixes `icw:` and `icw:log:` they are distinct. Seventeen of them name equations of the six shared blocks, which are printed once, in Part I: `eq:tree`, `eq:increments`, `eq:M`, `eq:tilt`, `eq:q`, `eq:endpoint`, `eq:gsum`, `eq:color`, `eq:h`, `eq:Sigma`, `eq:LLT`, `eq:loops`, `eq:resolvent`, `eq:Fres`, `eq:abelian`, `eq:5bound`, `eq:powerlog`; Part II's citations of them point to Part I. Six are printed in both Parts with the same content (`eq:avoid`, `eq:constants`, `eq:dual`, `eq:reverse`, `eq:seed`, `eq:derivative-limit`); (88) writes ρ where (2) writes ρ_0 . Two name different mathematics:

Delivered name	Part I (Report 125)	Part II (Report 123)
<code>thm:main</code>	Theorem 1.1: the leading equivalent and the killed local limit	Theorem 15.1: the logarithmic law, the circle, non-D-finiteness, the inverse
<code>eq:harmonic</code>	(56): discrete harmonicity of V	(100): the Brownian harmonic-measure series H_f

Symbols.

Symbol	Meanings
A, W	Part I: A the explicit whitening matrix (3); A_d the steps; A_0 a tail constant (Section 3.1); W a Brownian motion (Section 4) and $\mathcal{W}$ the wedge. Part II: W any whitening matrix with $W\Sigma W^\top = I$ (Section 17.3); A, B compact sets (Lemma 18.1); A also a jump-cutoff constant in Lemma 20.1. <i>Tempting false reading:</i> Part II's $W(1, 1)$ is a whitened vector, not a Brownian value; Part I's A is one admissible choice of Part II's W .
T	The killed operator on $\ell^\infty(\mathcal{D})$ (Section 12); $T = \lfloor n^{3/4} \rfloor$ (Section 8); in Part II the time constants $T(q)$, T of Section 19.
q	The tilted transition law q_{cd} (both); $q = p - 3$ in (43); the Hölder exponents q_0, q'_0 (Section 7); $q \in E$ in (86); the heat kernel q_t (Section 5); the transform variable of $\mathcal{L}(q)$ (26). Part II: $q > 1$ the annulus ratio and the outer radius in (100).
v	A displacement (both); the row functional v (Section 12); the inverse centre $v(y)$ (11); $v = Az/\sqrt{n}$ (Section 9); the drift v of (25); v_j a step (Part II).
L, λ	$L = \log y$ (Corollary 1.2, Section 24); $L(u)$ the Fourier matrix; L a return length (Section 4); in Part II also the multiplier of $\log n$ in Lemma 20.1. $\lambda = \log 9$ in the inverse; $\lambda(u)$ the Perron eigenvalue near $u = 0$; λ a small real tilt (Section 9); λ the variable of the Perron polynomial $D(\lambda, x, y)$.
h, H	h the maximum of an avoider (Section 2.1); h_c, h_c^* the correctors; $h = n^{1/3}$ (Section 8). H : in Part I a cycle supremum (Section 4), a radius (Section 6), a competing harmonic function and a seed length (Section 7), H_ϵ (Section 9); in Part II the series $H(z, x)$ (Section 16.2), the harmonic measure H_f (100) and a seed length.
B, K, J	B_P, B_Q, B_c^* the second-order correctors (46); $B : H$ the Frobenius product; B_l (Section 12); a compact set B (Section 9). Part II: $B = P + Q$ and the branch term B_0 (94); the cutoff B_n (Lemma 20.1). K_n the killed kernel (both); $K(z, x, y)$ the kernel polynomial (93); K an error power (Lemma 20.1); K_q in (86). J_τ the last-jump maximum (Section 6); in Part II $J(z, y)$ a series and J an angular interval (Section 19).
P, Q, D	Colours (both); in Section 16.2 also the series $P(z, x, y)$, $Q(z, x, y)$. D_j the descent steps; $D(\lambda, x, y)$ the Perron polynomial; $D(L)$ an envelope (79); D_M (28); $\mathcal{D}$ the reachable states (Section 12).

Symbol	Meanings
$\rho, \varrho, \varepsilon$	$\rho_0 = 1/9$ (2), but $\rho = 1/9$ in Section 12 and in Part II (88); ϱ the meander density (33). ε several small numbers, $\varepsilon(y)$ the inverse envelope; $\epsilon_l = 9^{-l}$.
r, e, M	$r = (2, 1)$, $r_P = 2$, $r_Q = 1$; $r = 1 + s $ (Section 6); $r_{\pm}(z)$ branch points and r_j radii (Part II). $e = (1, 0, P)$ the seed endpoint (both); e a coupling error (Sections 6, 8). $M(x, y)$ the step matrix; M_r the stopped radial maximum; M a power in tail bounds.

Against a377922-corner-polyhedra-schnyder. Its Part III writes $\nu = \pi/\theta$ for the exponent p here, E for the phase set ($\{P, Q\}$ here), $F(t)$ for the Fourier matrix ($L(u)$ here), $W = \Sigma^{-1/2}$ for a whitening, $\rho = \Sigma_{12}/\sqrt{\Sigma_{11}\Sigma_{22}}$ for a correlation ($-2/\sqrt{7}$ here, so its $\theta = \arccos(-\rho)$ is the θ here), and a, b for endpoint phases. Its κ is a tail rate in its jump bound, and its κ_P, κ_S are *amplitudes*; here $\kappa = 1 + p$ is the *exponent*, the analogue of its α_P, α_S . Its J is the angular interval of (H5), as in Part II.

Provenance and merge decisions

Two manuscripts of the session bundle of Reports 1–243, both dated 2 October 2026, with the header lines “REPORT 125” and “REPORT 123”; neither names an author or a tool, the PDF author fields are empty, and neither carries “prepared for private review” wording. They arrived in commit 60f54ea06 and were placed by 612787fb4 (batch 104). Neither pins a ProveIt commit or names a repository path, and neither cites the other in its text; but Report 125’s finite-check package contains a byte-for-byte copy of Report 123’s (its `checks/legacy/`, anchored by the manifest SHA-256 79f5dfa1...), and its checker reruns Report 123’s checks.

- **Part I**, bundle Report 125, archive A279571_Leading_Equivalent_and_Harmonic_Constant_Source.zip (952,684 bytes, 29 files; `report125.tex`, 1,575 lines, 25 pages). The base: its `report125.tex` became this `article.tex`. Contributes the leading equivalent with the characterization of C_A , the strong approximation at original times, the explicit second-order colour correctors, the normalized killed harmonic functions, the deep-entrance sharp survival, the killed local limit with last-window smoothing and midpoint reversal, the global endpoint bound, and the inverse bracket.
- **Part II**, bundle Report 123, archive A279571_Exact_Asymptotic_Exponent_and_Non_D_Finiteness_Source.zip (887,643 bytes, 23 files; `report123.tex`, 979 lines, 16 pages). Contributes the functional equations and the kernel, the logarithmic law by the functional limit theorem, the interior bridge, uniform annuli and exact-time gluing, the first proofs of the circle statement and of non-D-finiteness (printed in Part I), and the logarithmic inverse.

Where the merge had to choose:

1. *Base and order.* Report 125 is the base and Part I, because its Theorem 1.1 and Corollaries 1.2 and 12.1 imply every statement of Report 123’s Theorem 1.1 under the same hypotheses: the two-sided bounds follow from (9), the circle and arithmetic statements are the same text, and the inverse (91) follows from (12), since $v(y) = \frac{\log y}{\log 9} + \frac{\kappa}{\log 9} \log \log y + O(1)$. Report 123, created 49 minutes earlier, keeps priority within the bundle for the exponent, the circle statement and non-D-finiteness, and sole credit for the kernel of its Section 2.2 and for the annulus route.
2. *Shared text printed once.* Report 123’s Sections 2.1, 3.1, 3.2, 4.2, 8 and 9 are Report 125’s

Sections 2.1, 3.1, 3.2, 9.1, 12 and 13, word for word except in three places: Report 125's Section 12 opens with the statement of Corollary 12.1, where Report 123 has none; in two sentences of Sections 12.2 and 13.2 Report 125 cites its equivalent (9) where Report 123 cites its logarithmic law (89); and one clause of Section 3.2 is worded differently (note in Section 17.2). The text is printed once, in Part I, in Report 125's wording; Part II keeps the six headings with pointers, and its citations of the seventeen equations of these blocks point to Part I.

3. *Overlaps kept.* Report 123's Section 1 restates the definitions of Report 125's Section 1 with one more sentence, and its Section 3.3 covers the ground of Report 125's Section 3.3 with a general whitening matrix and different wording; both are printed in full in Part II, under the union rule.
4. *Second route.* Report 123's proof of its Theorem 1.1 is printed in full (Sections 18–21 and 24), with dated notes marking what Part I supersedes; its “What is not asserted” paragraph and its Section 10 describe Report 123 alone and are printed as delivered with dated notes.
5. *Front matter and bibliography.* The two title blocks and abstracts are printed at the head of their Parts; one table of contents replaces two. The bibliographies are merged: the entries `oeis`, `bb` and `fr` are identical in both, and Report 123's `whitt` is added to Report 125's five. Report 123's first section heading carried a manual `\hspace{1em}` on top of the delivered section-number spacing; it is dropped.
6. *Write additions.* Remark 26.1 with its proof, Proposition 26.2 with its proof, Corollary 26.3 with its proof and Remark 26.4 (Section 26); the two “Further questions and research” sections; the dated notes; the section labels added to delivered headings; and this front matter. Everything else is delivered text.

Relation to neighbouring reports and to the formal projects

All reports named here are in the same collection directory, `SetTheory/Cardinals/docs/reports/generating-functions-and-asymptotics/oeis-sequence-asymptotics/`.

The cone theorem of a377922-corner-polyhedra-schnyder, Part III (bundle Report 112). Its Section 22.5 (`cps:log:sec:siblings`) lists Report 123 among three bundle reports that run one argument: a Perron tilt to a centred walk with a finite phase, a bounded Poisson corrector and the martingale functional limit theorem, a Fourier local limit theorem with an edge-phase aperiodicity argument, an exact-time interior bridge, Brownian wedge harmonic measure over geometric annuli, two first-hit prefixes glued at the exact time, non-D-finiteness by a positive derivative and the G -function theorem, and an $o(\log \log y)$ inverse. That is the whole of Part II's proof, and Report 112 states it as Theorem 24.1 there (`cps:log:thm:cone`), for Markov-additive walks with finitely many phases and a uniform exponential moment, under hypotheses (H1)–(H5). Report 123's walk satisfies (H1)–(H4), but *not* (H5) at the endpoints the theorem states, which lie at the spatial origin: no legal path from o returns to the origin, the dual has no seed there, and $K_n(o, ((0,0), b)) = 0$ for $n \geq 1$ and both colours b (Remark 26.1). Report 123's exponent is therefore an instance of that theorem's *proof*, run at the endpoint $e = (1, 0, P)$, not of its statement. Proposition 26.2 writes this out as a statement, Theorem 24.1 at arbitrary fixed endpoint states with seeds, and Corollary 26.3 derives Report 123's two-sided bounds from it. A279571 is a concrete example for Question Q3 there (`cps:log:q:seeds`, endpoints without seeds): at the origin endpoints there is no exponent

at all. For this one walk Part I proves the kind of statement that Question Q1 there asks for (`cps:log:q:amplitude`, an amplitude form of Theorem 24.1), at every fixed pair of states and without a cycle buffer, by a phase-sensitive killed local limit, the route that question names as missing. The walk is not an instance of that question, whose hypotheses include (H5) at the origin; Part I needs no (H5), and the positivity of V and V^* at the chosen endpoints takes its place. Its argument uses $4 < p < 5$, so it does not generalize as written (Remark 26.4). It is a third route to amplitudes beside Part II there (comparison with an iid cone walk through a fixed cycle buffer) and Part III there (annuli, logarithmic scale only). At the write, Section 22.5 there and its README still call Report 123 unplaced; a reciprocal note is proposed for a separate commit.

Inverses and non-D-finiteness. Corollary 1.2 follows from Theorem 20.1 of `a377922-corner-polyhedra-schnyder` (`cps:amp:thm:inverse`) with $\gamma = C_A$, $\mu = 9$, $\alpha = \kappa$: that theorem's centre $r(Y)$ differs from $v(y)$ by $o(1)$ (its display (20.2), `cps:amp:eq:inverseexpansion`), so its bracket, widened by $|r-v|$, is (12). That theorem needs no monotonicity of a_n ; Report 125's proof (Section 11) uses it for the indices below $m(L)$. Its mechanics are those of the transseries volume (`Analysis/Transseries/docs/series-and-transseries/Transseries_And_Inversion/transseries_and_inversion.tex`): `p0:thm:lambert-core` (branch W_{-1}), `p0:thm:lambert-centered` and `p0:thm:staircase`; $N(y)$ is the integer staircase of `p0:def:three-inverses`. No novelty is claimed for the inversion mechanics. The non-D-finiteness argument of Section 13 is an instance of the general criterion of Remark 27.1 of `a377922-corner-polyhedra-schnyder` (`cps:log:rem:criterion`) and of Remark 10.1 of `a348351-one-sided-rectangulations` (`osr:rem:criterion`), with derivative order $5 > \kappa - 1$; the radius $1/9$ is rational, so the algebraicity point added to that criterion at its independent check does not arise.

`a348351-one-sided-rectangulations` (bundle Report 111) runs the same spine as Part II for a four-colour walk with bounded steps and its endpoint at the origin, where its seeds exist (its Remark 1.2, `osr:rem:cone`). Its Questions 1 and 2 (`osr:q:equivalent`, `osr:q:harmonic`) name Report 125 as a model for the full equivalent and the harmonic functions “to be filed by a later intake batch”; it is Part I here. Whether Part I's route carries over there was not checked; as written it does not, since there $p = \alpha - 1 \approx 1.957$, while Part I uses $p > 2$ in (39) and $p > 4$ in Section 7.2. Its uncertified observation $a(n)\Gamma^{-n}n^\alpha \approx 4.65$ at $n = 599$ is of the same kind as the one recorded in Section 14.1.

Other Britt–Beaton classes. Batch 104 placed two further inversion-sequence reports from the same bundle beside this one, `a279544-inversion-kernel-classes` (A279544, A279567, A279569, A279558; bundle Reports 118, 119, 122: algebraic kernels, square-root singularities) and `a279551-inversion-log-deficit` (A279551, A279556; bundle Reports 126, 127, 129: stretched-exponential deficits). They share with this report only the definition of inversion sequences and Britt and Beaton's paper: no theorem, method, notation or text, and no cross-citation in either direction. Other reports of the collection cite inversion-sequence papers (`a113227-powered-catalan`, `a336070-weak-ascents`, `a196275-column-convex-permutominoes`) but treat no Britt–Beaton class.

Formal projects. No Lean or Rocq development of the repository touches A279571, inversion sequences or Markov-additive cone walks; no statement of this report is formalized, and its placement confers no formal status.

Part I

The leading equivalent and a harmonic constant

Bundle Report 125, archive A279571_Leading_Equivalent_and_Harmonic_Constant_Source.zip, delivered as `report125.tex` (25 pages), dated 2 October 2026. Delivered title “A279571: the leading equivalent”, subtitle “A positive harmonic constant and a sharp threshold inverse”, with the header “REPORT 125” and the line “Standalone proof and exact finite-check companion”. Printed as delivered, with `[write]` notes; section, statement and equation numbers are those of the delivery. Its Sections 2.1, 3.1, 3.2, 9.1, 12 and 13 are also Report 123’s Sections 2.1, 3.1, 3.2, 4.2, 8 and 9 (Part II). Section 14.1 is added in the write.

Abstract of this Part (as delivered). Let a_n count inversion sequences avoiding 100, 101, and 201. We prove

$$a_n \sim C_A 9^n n^{-\kappa}, \quad \kappa = 1 + \frac{\pi}{\arccos(2/\sqrt{7})}, \quad 0 < C_A < \infty.$$

The constant is characterized by normalized forward and reversed killed harmonic functions and an absolutely convergent positive endpoint series. The proof treats the exact two-color walk at its original integer times. A strong approximation for full regenerative cycle paths supplies uniform rough bounds. Explicit first- and second-order color correctors construct the harmonic functions. A deep entrance argument, a side-uniform Brownian limit, last-window smoothing, midpoint reversal, and a global endpoint bound give the required killed local limit and justify the endpoint sum. The threshold inverse has an explicit real center and a ceiling bracket whose pre-rounding width tends to zero. We also prove uniqueness of the convergence-circle singularity and non-D-finiteness. No amplitude digits, quantitative remainder rate, effective onset, or correction expansion is asserted.

Reading conventions. This Part’s letters are the reference letters of the report. Here A is the explicit whitening matrix (3), $\rho_0 = 1/9$ in (2) but $\rho = 1/9$ in Section 12, and $q, v, L, \lambda, h, H, T, D$ carry several meanings (front matter, “Notation across the two Parts”). Part II writes W for a whitening matrix and ρ for $1/9$.

1 Definitions, results, and normalization

An inversion sequence of length n is an integer sequence $(e_1, \dots, e_n)$ with $0 \leq e_i < i$. The three forbidden patterns are exactly the triples $i < j < k$ satisfying

$$e_i > e_j, \quad e_j \leq e_k, \quad e_i \geq e_k. \quad (1)$$

Let a_n count the avoiders, with $a_0 = 1$, and write $F(z) = \sum_{n \geq 0} a_n z^n$. This is OEIS A279571 and Class 2106 of Britt and Beaton [1, 2]. Its initial terms are

$$1, 1, 2, 6, 22, 92, 424, 2106, 11102, 61436, 353980, 2110366, \dots$$

All logarithms are natural. Define

$$\theta = \arccos \frac{2}{\sqrt{7}}, \quad p = \frac{\pi}{\theta}, \quad \kappa = p + 1, \quad \rho_0 = \frac{1}{9}. \quad (2)$$

The exact walk below has colors P, Q , positions in $\mathcal{Q} = \mathbb{Z}_{\geq 0}^2$, start $o = (0, 0, P)$, stationary color masses $\pi = (2/3, 1/3)$, and covariance

$$\Sigma = \begin{pmatrix} 7/3 & -10/3 \\ -10/3 & 25/3 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 2/5 \\ 0 & \sqrt{3}/5 \end{pmatrix}, \quad A\Sigma A^\top = I. \quad (3)$$

The wedge $\mathcal{W} = A\mathbb{R}_{\geq 0}^2$ has angles $0 < \phi < \theta$. Set

$$u(w) = |w|^p \sin(p \arg w), \quad U(s) = u(As), \quad b_\theta = \frac{2^{1-p}}{\theta \Gamma(p+1)}. \quad (4)$$

Values on the wedge sides are zero. This fixes the multiplicative normalization completely; equivalently, on the quadrant,

$$U(a, b) = \operatorname{Im} \left(a + \frac{2 + i\sqrt{3}}{5} b \right)^p$$

with the branch of argument between 0 and θ .

Theorem 1.1 (Leading equivalent and killed local limit). *Let K_n be the exact tilted walk killed on leaving $\mathcal{Q}$, and let K_n^* be its stationary-color time reversal, also killed on $\mathcal{Q}$. Writing τ, τ^* for their exit times, the limits*

$$V(s, c) = \lim_{n \rightarrow \infty} \mathbb{E}_{s,c}[U(S_n); \tau > n], \quad V^*(t, d) = \lim_{n \rightarrow \infty} \mathbb{E}_{t,d}^*[U(S_n); \tau^* > n] \quad (5)$$

exist, are nonnegative killed-harmonic functions, and satisfy, for every $\delta > 0$,

$$V(s, c) = U(s) + O_\delta((1 + |s|)^{p-1+\delta}), \quad (6)$$

and the same bound for V^ . For fixed states $(s, c), (t, d)$,*

$$K_n((s, c), (t, d)) = \frac{b_\theta \pi_d}{\sqrt{\det \Sigma}} V(s, c) V^*(t, d) n^{-p-1} + o(n^{-p-1}). \quad (7)$$

Uniformly over all $(t, d) \in \mathcal{Q} \times \{P, Q\}$ and $n \geq 1$,

$$K_n(o, (t, d)) \leq C(1 + |t|)^{2p} n^{-p-1}. \quad (8)$$

Consequently

$$a_n \sim C_A 9^n n^{-\kappa}, \quad (9)$$

where, with $r_P = 2, r_Q = 1$ and $g((a, b), d) = 3^{-a}(3/5)^b/r_d$,

$$C_A = \frac{2}{9} \frac{b_\theta}{\sqrt{\det \Sigma}} V(o) \sum_{t \in \mathcal{Q}} \sum_{d=P,Q} \pi_d g(t, d) V^*(t, d). \quad (10)$$

The series is absolutely convergent, and $0 < C_A < \infty$.

[write] **5 October 2026.** Part II, Theorem 15.1, proves the weaker two-sided bounds $c_\varepsilon 9^n n^{-\kappa-\varepsilon} \leq a_n \leq C_\varepsilon 9^n n^{-\kappa+\varepsilon}$ by another route (Sections 18–21); (9) implies them. Display (7) is, for this walk and every fixed pair of states, the amplitude form that Question Q1 of [a377922-corner-polyhedra-schnyder](#) (`cps:log:q:amplitude`) asks for its general cone theorem, although this walk is not an instance of that question, since its hypothesis (H5) fails at the origin (Remarks 26.1 and 26.4). At the origin endpoints $V^*((0, 0), b) = 0$ and $K_n(o, ((0, 0), b)) = 0$ for $n \geq 1$ (Remark 26.1), which is consistent with (7). *Observation, not a claim:* on the exact OEIS terms (the shipped b-file), $a_n 9^{-n} n^\kappa = 77.91, 83.24, 86.55, 87.65$ at $n = 100, 200, 500, 1000$ (double precision), still increasing; no digit of C_A is certified (Section 14.1, items 1–2).

Corollary 1.2 (Threshold inverse). *For $N(y) = \min\{n \geq 0 : a_n \geq y\}$, put $\lambda = \log 9$, $L = \log y$, and*

$$v(y) = \frac{L + \kappa \log L - \kappa \log \lambda - \log C_A}{\lambda}. \quad (11)$$

There is a positive function $\varepsilon(y) \rightarrow 0$ such that, for all sufficiently large y ,

$$\lceil v(y) - \varepsilon(y) \rceil \leq N(y) \leq \lceil v(y) + \varepsilon(y) \rceil. \quad (12)$$

In particular $N(y) = v(y) + O(1)$. The exact identity $N(y) = \lceil v(y) \rceil$ follows whenever the displayed real interval contains no integer; no unconditional ceiling identity is claimed.

[write] **5 October 2026.** Corollary 1.2 follows from Theorem 20.1 of `a377922-corner-polyhedra-sch nyder (cps:amp:thm:inverse)` with $\gamma = C_A$, $\mu = 9$, $\alpha = \kappa$, and is an instance of the transseries volume’s Lambert- W inversion (front matter, “Relation to neighbouring reports”). Part II’s inverse (91) is weaker: since $v(y) = \frac{\log y}{\log 9} + \frac{\kappa}{\log 9} \log \log y + O(1)$, the bracket (12) gives (91) with error $O(1)$ in place of $o(\log \log y)$.

The numbers p, κ are irrational, with $4 < p < 5$. We also prove that F is not D-finite over $\mathbb{C}(z)$ and that its only singularity on its convergence circle is $1/9$; see Section 12.

Scope and attribution. The generating tree is the exact combinatorial input of Britt–Beaton [2, Section 2.5, equation (2.44)]. Their Section 4.2 and Appendix B.3, and the OEIS formula credited to Václav Kotěšovec on 13 January 2026 after Nicholas R. Beaton, contain prior numerical exponent and amplitude estimates. Those estimates are credited rather than presented as new results. No numerical fitting is used here. The constant in (10) is a mathematical characterization, without certified decimal digits or effective error bounds. The leading equivalent excludes a nonconvergent bounded amplitude modulation, but gives no correction series or complex local expansion.

The proof does not apply an iid cone theorem to the colored walk or kill only at color returns. The external probabilistic inputs are a full-cycle original-time strong approximation and the Brownian wedge heat kernel. All discrete stopping, correction, local smoothing, and endpoint-summation arguments needed for Theorem 1.1 are given below.

2 From the generating tree to a quadrant walk

[write] **5 October 2026.** Sections 2.1, 3.1 and 3.2 are also Report 123’s Sections 2.1, 3.1 and 3.2, word for word except for one clause of Section 3.2 (note in Part II, Section 17.2). Report 123’s Section 2.2, the functional equations and the kernel of the generating tree, is printed in Part II, Section 16.2; it is not in this Part.

2.1 Exact combinatorial coordinates

For a nonempty avoider let h be its maximum, and let its color be P when its last entry equals h , and Q otherwise. Let k be the number of values below its last entry that may legally be appended. The source labels are $(n, h, k)_P$ and $(n, h, k)_Q$. Its exact rule is

$$\begin{aligned} (n, h, k)_P &\longrightarrow (n+1, i, k+i-h)_P & (h \leq i \leq n), \\ (n, h, k)_Q &\longrightarrow (n+1, i, k+i-h-1)_P & (h+1 \leq i \leq n), \\ (n, h, k)_c &\longrightarrow (n+1, h, i)_Q & (0 \leq i < k; \ c = P, Q). \end{aligned}$$

To see the descent rule directly, list the valid smaller values as $j_0 < \dots < j_{k-1}$. Choosing j_i leaves exactly its i smaller valid values: repeating j_i would form 100, and returning to a larger listed value would form 201 with the earlier maximum. Smaller values create no new forbidden triple, and a previously forbidden value cannot become valid by appending another entry. From color Q , repetition is forbidden by 100, returning to the old maximum by 101, and any intermediate ascent by 201. A new maximum creates fresh permitted values above the old maximum. Thus the first two lines differ precisely in whether the old maximum remains available as a future descent, which explains their endpoints and the extra decrement in the Q rule.

Put $a = n - h$ and $b = k$, keeping the empty root $(0, 0, P)$. Reindex the new maximum by $d = i - h$, and the descent rank by $j = k - i$. The rules become

$$\begin{aligned} P(a, b) &\longrightarrow P(a + 1 - d, b + d) & (0 \leq d \leq a), \\ Q(a, b) &\longrightarrow P(a + 1 - d, b + d - 1) & (1 \leq d \leq a), \\ c(a, b) &\longrightarrow Q(a + 1, b - j) & (1 \leq j \leq b; c = P, Q). \end{aligned} \tag{13}$$

The sole size-one object has coordinates $(1, 0, P)$. From it onward replace a by $a - 1$. Then every edge is a translation-invariant displacement

$$A_d = (1 - d, d)(P \rightarrow P, d \geq 0), \quad C_d = (1 - d, d - 1)(Q \rightarrow P, d \geq 1), \quad D_j = (1, -j)(c \rightarrow Q, j \geq 1), \tag{14}$$

retained exactly when the endpoint lies in $\mathcal{Q} = \mathbb{Z}_{\geq 0}^2$. The shifted starting state is $o = (0, 0, P)$. For example, $a + 1 - d \geq 0$ is precisely the original bound $d \leq a + 1$ after the shift. No additional boundary is missing: the sum of the shifted coordinates begins at zero and every step increases it by at most one. After $n - 1$ steps it is therefore at most $n - 1$ automatically. Consequently a_n , for $n \geq 1$, is the number of retained paths of length $n - 1$ from o , summed over endpoints and colors.

3 A positive change of weights and its covariance

3.1 The Perron tilt and endpoint identity

The unrestricted step matrix, with row denoting the old color, is

$$M(x, y) = \begin{pmatrix} x^2/(x - y) & x/(y - 1) \\ x/(x - y) & x/(y - 1) \end{pmatrix}, \quad x > y > 1. \tag{15}$$

For example its first entry is $\sum_{d \geq 0} x^{1-d} y^d = x^2/(x - y)$. At $(x, y) = (3, 5/3)$,

$$M = \begin{pmatrix} 27/4 & 9/2 \\ 9/4 & 9/2 \end{pmatrix}, \quad Mr = 9r, \quad r = \begin{pmatrix} 2 \\ 1 \end{pmatrix}.$$

The other eigenvalue is $9/4$, so this is the positive simple Perron root. Give each individual edge $c \rightarrow d$ with displacement v probability

$$q_{cd}(v) = \frac{3^{v_1} (5/3)^{v_2} r_d}{9r_c}. \tag{16}$$

The row sums equal one. Explicitly,

$$\begin{aligned} q_{PP}(A_d) &= \frac{1}{3} (5/9)^d & (d \geq 0), \\ q_{QP}(C_d) &= \frac{2}{9} (5/9)^{d-1} & (d \geq 1), \\ q_{PQ}(D_j) &= \frac{1}{6} (3/5)^j, \quad q_{QQ}(D_j) = \frac{1}{3} (3/5)^j & (j \geq 1). \end{aligned} \tag{17}$$

Hence there are fixed $A_0, b_0 > 0$ such that, from either color, $\mathbb{P}(|v| > u) \leq A_0 e^{-b_0 u}$. The same statement holds in any fixed linearly transformed norm.

Let $K_m((s, c), (t, d))$ be the m -step kernel of this process killed on leaving $\mathcal{Q}$. Telescoping (16) along a path yields

$$a_n = 2 \cdot 9^{n-1} \sum_{t=(a,b) \in \mathcal{Q}} \sum_{d=P,Q} K_{n-1}(o, (t, d)) g(t, d), \quad g((a, b), d) = \frac{3^{-a}(3/5)^b}{r_d}. \quad (18)$$

All terms are nonnegative. In particular, $0 < g \leq 1$, and

$$\sum_{t,d} g(t, d) = (1/2 + 1) \frac{1}{1 - 1/3} \frac{1}{1 - 3/5} = \frac{45}{8}. \quad (19)$$

Its tails are exponentially small. Equation (18) also gives $a_n \leq 2 \cdot 9^{n-1}$ for $n \geq 1$. The exponentially summable endpoint weight, rather than unweighted survival, is essential to the final power.

3.2 The color corrector

Write (S_j, C_j) for the unrestricted position and color process. Summing (17) gives

$$P_0 = \begin{pmatrix} 3/4 & 1/4 \\ 1/2 & 1/2 \end{pmatrix}, \quad \pi = (2/3, 1/3). \quad (20)$$

This matrix is primitive. The conditional means of v are

$$m_P = (1/16, 5/16), \quad m_Q = (-1/8, -5/8), \quad \pi_P m_P + \pi_Q m_Q = 0.$$

The vectors

$$h_P = (1/12, 5/12), \quad h_Q = (-1/6, -5/6) \quad (21)$$

satisfy $\pi h = 0$ and $(I - P_0)h = m$. Thus

$$Z_j = S_j - S_0 + h(C_j) - h(C_0)$$

is a square-integrable martingale. Its increment on an edge $c \rightarrow d$ is $v + h_d - h_c$; its mean conditional on the initial color c is zero. To compute its covariance one may use

$$\sum_{k \geq 0} r^k = \frac{1}{1-r}, \quad \sum_{k \geq 0} k r^k = \frac{r}{(1-r)^2}, \quad \sum_{k \geq 0} k^2 r^k = \frac{r(1+r)}{(1-r)^3}$$

in (17). The two conditional covariance matrices are

$$\Gamma_P = \frac{1}{64} \begin{pmatrix} 147 & -195 \\ -195 & 495 \end{pmatrix}, \quad \Gamma_Q = \frac{1}{32} \begin{pmatrix} 77 & -125 \\ -125 & 305 \end{pmatrix}.$$

Consequently the effective covariance is

$$\Sigma = \pi_P \Gamma_P + \pi_Q \Gamma_Q = \begin{pmatrix} 7/3 & -10/3 \\ -10/3 & 25/3 \end{pmatrix}, \quad \det \Sigma = 25/3 > 0. \quad (22)$$

This includes the serial dependence induced by colors; using only the uncorrected single-step covariance would be invalid.

A second exact derivation is useful. Clearing the characteristic polynomial of (15) gives

$$D(\lambda, x, y) = \lambda^2(x - y)(y - 1) - \lambda\{x^2(y - 1) + x(x - y)\} + x^2(x - 1).$$

At $(\lambda, x, y) = (9, 3, 5/3)$ one has $D = D_x = D_y = 0$, $D_\lambda = 6$, and $(D_{xx}, D_{xy}, D_{yy}) = (-14, 36, -162)$. Implicit differentiation of the simple Perron root gives

$$\partial_{\log x_i} \partial_{\log x_j} \log \lambda = -\frac{x_i x_j D_{x_i x_j}}{9 D_\lambda} = \Sigma_{ij}.$$

The first derivatives vanish, which explains the absence of extra terms.

3.3 Explicit whitening and reversed correctors

The matrix A of (3) satisfies

$$A^\top A = \Sigma^{-1} = \begin{pmatrix} 1 & 2/5 \\ 2/5 & 7/25 \end{pmatrix}, \quad \det A = \frac{\sqrt{3}}{5} = \frac{1}{\sqrt{\det \Sigma}}.$$

Its boundary vectors have scalar-product cosine $2/\sqrt{7}$, proving that the opening angle is exactly (2). In the physical quadrant, $\Sigma : D^2 U = 0$ away from the origin. Here $B : H$ denotes $\sum_{i,j} B_{ij} H_{ij}$.

The reversed free kernel is

$$q_{dc}^*(-v) = \frac{\pi_c}{\pi_d} q_{cd}(v). \quad (23)$$

Its row sums are one by stationarity. Its color matrix is again P_0 . It has uniform exponential jump tails, stationary zero drift, and the same covariance per original time: a stationary reversed displacement is the negative of a forward displacement. More explicitly,

$$\begin{aligned} m_P^* &= (1/2, -5/4), & m_Q^* &= (-1, 5/2), \\ h_P^* &= (2/3, -5/3), & h_Q^* &= (-4/3, 10/3), \\ \Gamma_P^* &= \begin{pmatrix} 3 & -15/4 \\ -15/4 & 15/2 \end{pmatrix}, & \Gamma_Q^* &= \begin{pmatrix} 1 & -5/2 \\ -5/2 & 10 \end{pmatrix}. \end{aligned}$$

Thus $(I - P_0)h^* = m^*$, $\pi h^* = 0$, and $\sum_c \pi_c \Gamma_c^* = \Sigma$. Reversal of every retained path gives

$$\pi_c K_n((s, c), (t, d)) = \pi_d K_n^*((t, d), (s, c)). \quad (24)$$

The invariant free measure is spatial counting measure times π . Killing at every original step preserves this reversal identity.

The auxiliary kernels are defined on both colors at every point of the quadrant. From o no forward path enters a Q state with first coordinate zero. In reversal every step out of such a Q state decreases the first coordinate by one, so it cannot reach the interior before dying. Such extra auxiliary states therefore do not introduce new counted paths. They may have zero harmonic value, which is allowed in the theorem.

4 A strong approximation at original times

4.1 The imported theorem and its exact hypotheses

We use Bashtova–Shashkin [3, Conditions (A), (B), Theorem 1, pp. 3–4]; the published theorem is Theorem 2.1. In the form needed here, a separable process $X(t)$ has successive regeneration times $0 = T_0 < T_1 < \dots$ such that the pairs

$$(T_j - T_{j-1}, (X(T_{j-1} + t) - X(T_{j-1}))_{0 \leq t \leq T_j - T_{j-1}})$$

are iid. Write $L = T_1$, $Z = X(T_1) - X(0)$, and $H = \sup_{0 \leq t \leq L} |X(t) - X(0)|$. Assume $\mathbb{E}e^{aL} + \mathbb{E}e^{aH} < \infty$ for some $a > 0$. Then, with $v = \mathbb{E}Z/\mathbb{E}L$ and $\Sigma_{\text{reg}} = \mathbb{E}[(Z - vL)(Z - vL)^\top]/\mathbb{E}L$, there is a coupling with covariance- Σ_{reg} Brownian motion W and constants $a_0, b_0, c_0 > 0$ such that

$$\mathbb{P}\left(\sup_{0 \leq t \leq N} |X(t) - X(0) - vt - W_t| > c_0 \log(N+2) + x\right) \leq a_0 e^{-b_0 x} \quad (N \geq 1, x \geq 0). \quad (25)$$

This theorem makes no independence assumption between L and Z and imposes no nonlattice condition. Its control is over the full original real-time path, not just the regeneration endpoints.

4.2 Verifying full-cycle moments and covariance

For an initial color c , take successive positive returns to that same color, and set $X(t) = S_{\lfloor t \rfloor} - S_0$. The piecewise-constant process is separable. Translation invariance and the strong Markov property make the entire cycle elements iid, with no delayed initial cycle. The finite color chain makes the return length L geometrically tailed. For example a P return has length one with probability $3/4$; otherwise the chain enters Q and returns with probability $1/2$ on each subsequent step. Thus $\mathbb{E}_P L = 3/2$, while $\mathbb{E}_Q L = 3$.

Given a color path, the displacements are independent with one of four transition-conditioned laws. Each has an exponential moment near zero. If $M(a)$ is the maximum of their moment generating functions for jump magnitude, then $M(a) \rightarrow 1$ as $a \downarrow 0$, and

$$\mathbb{E}e^{aH} \leq \mathbb{E}e^{a \sum_{j=1}^L |\Delta S_j|} \leq \mathbb{E}M(a)^L < \infty$$

for small enough a . This verifies the within-cycle maximum hypothesis, which cannot be replaced by an endpoint-only moment assumption.

Stop $S + h(C)$ at the return to c . Exponential cycle moments justify optional stopping and make the color offsets cancel, so $\mathbb{E}_c Z = 0$. The stopped martingale isometry and the stationary occupation formula give

$$\mathbb{E}_c[ZZ^\top] = \sum_j \frac{\pi_j}{\pi_c} \Gamma_j, \quad \mathbb{E}_c L = \frac{1}{\pi_c}, \quad \frac{\mathbb{E}_c[ZZ^\top]}{\mathbb{E}_c L} = \Sigma.$$

For an independent explicit calculation, let $\mathcal{L}(q)_{cd} = \sum_v q_{cd}(v)e^{q \cdot v}$. A first P return satisfies

$$\mathbb{E}[z^L e^{q \cdot Z}] = z \mathcal{L}_{PP}(q) + \frac{z^2 \mathcal{L}_{PQ}(q) \mathcal{L}_{QP}(q)}{1 - z \mathcal{L}_{QQ}(q)}. \quad (26)$$

Differentiating at $z = 1, q = 0$ gives

$$\mathbb{E}(Z_1, Z_2, L) = (0, 0, 3/2), \quad \text{Cov}(Z_1, Z_2, L) = \begin{pmatrix} 7/2 & -5 & 7/8 \\ -5 & 25/2 & -25/8 \\ 7/8 & -25/8 & 5/4 \end{pmatrix}, \quad (27)$$

whose determinant is $225/32$. Displacement and length are correlated; the correlation is harmless here because $v = 0$ in the regenerative covariance formula, rather than because it vanishes.

The reversed transition-conditioned laws are negatives of the forward ones with the color pair interchanged, so all preceding cycle-moment arguments hold in reversal. Use the displayed h^*, Γ^* to obtain its covariance in the same way. Taking maxima of constants over two initial colors and two time directions proves that both free chains have (25) with $v = 0$ and covariance Σ , uniformly in the initial position and color. In particular, for every $M > 0$, a suitable D_M gives

$$\mathbb{P} \left(\sup_{0 \leq t \leq n} |S_{[t]} - s - W_t| > D_M \log(n+2) \right) \leq C_M (n+2)^{-M}. \quad (28)$$

Every later use restarts a free coupling at a specified state. No adaptedness to a future random entrance time is asserted. Original-time killing is compared on the whole coupled path.

5 Brownian wedge normalization and side uniformity

For standard Brownian motion in $\mathcal{W}$, let $q_t(w, z)$ be the killed transition density with respect to two-dimensional Lebesgue measure, and τ_B its exit time. Bañuelos–Smits [4, Lemma 1, equation (2.2)], with generator $\frac{1}{2}\Delta$ and angular normalization $\sqrt{2/\theta}$, gives

$$q_t(w, z) = \frac{2}{\theta t} e^{-(|w|^2 + |z|^2)/(2t)} \sum_{j \geq 1} I_{jp}(|w||z|/t) \sin(jp \arg w) \sin(jp \arg z). \quad (29)$$

The leading Bessel term $I_p(a) \sim (a/2)^p / \Gamma(p+1)$ yields b_θ in (4). We now give the uniformity needed when the starting angle approaches either side.

By scaling take $t = 1$ and write $w = (r_0, \alpha)$, $z = (r, \beta)$ in polar coordinates. For $0 < \alpha < \theta$,

$$|\sin(jp\alpha)| \leq j \sin(p\alpha), \quad I_\nu(t) \leq (t/2)^\nu e^t / \Gamma(\nu+1) \quad (\nu \geq 0).$$

The latter follows directly from the Bessel series. To bound the sum, put $G(t) = \sum_{j \geq 1} j t^{jp} / \Gamma(jp+1)$. For $t \leq 1$, $G(t) \leq C t^p$. For $t \geq 1$, put $m = \lfloor jp \rfloor$. Distinct j give distinct m , $j \leq m$, $t^{jp} \leq t^{m+1}$, and $\Gamma(jp+1) \geq cm!$. Therefore

$$G(t) \leq C t \sum_{m \geq 0} \frac{m t^m}{m!} \leq C t^2 e^t, \quad G(t) \leq C t^p (1+t)^2 e^t \quad (t > 0).$$

After division of (29) by $u(w) = r_0^p \sin(p\alpha)$, these inequalities give, for $0 < r_0 \leq 1$, the angle-independent integrable envelope

$$C r^p (1+r)^2 e^{-r^2/2+2r}. \quad (30)$$

It is integrable with the radial Jacobian $r dr d\beta$. The $j = 1$ term tends to $b_\theta u(z) e^{-|z|^2/2}$ after this division, uniformly in α . All $j \geq 2$ terms tend to zero, also uniformly in α , by the sine bound and the convergent Bessel series. Dominated convergence now proves the stronger L^1 limit

$$\sup_{0 < \alpha < \theta} \int_{\mathcal{W}} \left| \frac{q_1((r_0, \alpha), z)}{u((r_0, \alpha))} - b_\theta u(z) e^{-|z|^2/2} \right| dz \rightarrow 0 \quad (r_0 \downarrow 0). \quad (31)$$

In particular, uniformly in all starting angles and $|w| = o(\sqrt{t})$,

$$\mathbb{P}_w(\tau_B > t) \sim k_\theta u(w) t^{-p/2}, \quad \text{Law}(W_t/\sqrt{t} \mid \tau_B > t) \implies \varrho(z) dz, \quad (32)$$

where

$$I_1 = \int_{\mathcal{W}} u(z) e^{-|z|^2/2} dz = \frac{2^{p/2+1}}{p} \Gamma(p/2 + 1), \quad k_\theta = b_\theta I_1, \quad \varrho(z) = \frac{u(z) e^{-|z|^2/2}}{I_1}. \quad (33)$$

The corresponding joint endpoint limit holds for every bounded test function with error $o(u(w)t^{-p/2})$ uniformly over these starts. The envelope for $|w| \leq \sqrt{t}$ and the trivial bound for larger w also give the useful global upper bound

$$\mathbb{P}_w(\tau_B > t) \leq C \min\{1, |w|^p t^{-p/2}\}. \quad (34)$$

At the sides the survival probability is zero. These statements require no angular lower bound away from the sides.

Two elementary normalization integrals will be used later:

$$I_2 = \int_{\mathcal{W}} u(w)^2 e^{-|w|^2} dw = \frac{\theta \Gamma(p+1)}{4}, \quad b_\theta^2 2^{p+1} I_2 = b_\theta. \quad (35)$$

For example $p = 2, \theta = \pi/2$ gives $u(x, y) = 2xy$ and $b_\theta = 1/(2\pi)$, matching the product of the two killed half-line Brownian kernels. This checks the Lebesgue-measure convention.

6 Uniform rough bounds and stopped moments

Every assertion in this section holds for both time directions and both initial colors. Constants may depend on a displayed exponent or δ , but not on the starting position s or the time. Write $r = 1 + |s|$. Norms before and after whitening are equivalent, with fixed constants.

6.1 Survival and free moments

A sup-norm coupling error e is absorbed in an upper survival comparison by starting covariance- Σ Brownian motion at $s + e(1, 1)$. If the lattice path stays in the quadrant, this shifted Brownian path does too, apart from the coupling-failure event. A harmless extra unit in e avoids equality on the sides. Equations (28) and (34) imply, for every $M > 0$,

$$\mathbb{P}_{s,c}(\tau > n) \leq C_M \min\{1, n^{-p/2}(r + \log(n+2))^p\} + C_M(n+2)^{-M}. \quad (36)$$

Integrating the exponential excess tail in (25), not merely its polynomial corollary, bounds every fixed moment of the coupling error by $C_a \log^a(n+2)$. Brownian maximal moments then give

$$\mathbb{E}_{s,c} \max_{j \leq n} (1 + |S_j|)^a \leq C_a (r + \sqrt{n})^a \quad (a > 0). \quad (37)$$

The same full coupling inequality and the Brownian Gaussian tail give, for each $M > 0$,

$$\mathbb{P}_{s,c}(|S_n - s| > \sqrt{n} \log(n+2)) \leq C_M(n+2)^{-M}. \quad (38)$$

The Brownian part is bounded by $C \exp(-c \log^2(n+2))$ and the coupling part by $C \exp(-c \sqrt{n} \log(n+2))$ for large n . Either tail is smaller than any prescribed power; finitely many small times are absorbed in the constant. Changing the fixed multiplier of $\sqrt{n} \log(n+2)$ has no effect on these conclusions. Spatial homogeneity makes the constants independent of s .

6.2 Exit time, radial maximum, and last jump

Define the stopped quantities

$$M_\tau = 1 + \max_{0 \leq k \leq \tau} |S_k|, \quad J_\tau = 1 + \max_{1 \leq k \leq \tau} |S_k - S_{k-1}|.$$

The exit location is included. Summing (36) and splitting at r^2 shows

$$\mathbb{E}_{s,c} \tau \leq Cr^2. \quad (39)$$

Indeed $p > 2$, and the tail is bounded by a multiple of $r^p \sum_{n > r^2} n^{-p/2} + \sum_{n > r^2} n^{-p/2} \log^p(n+2)$, which is $O(r^2)$. In particular $\tau < \infty$ almost surely.

For $H \geq 2r$, take $N = \lfloor H^2/(D \log H) \rfloor$, adjusting bounded H by constants. If $M_\tau > H$, either $\tau > N$ or the free maximum up to N exceeds H . Brownian maximal tails and the exponential coupling inequality make the latter at most $CH^{-M'}$ for any chosen M' , by choosing D sufficiently large. The survival term in (36) gives

$$\mathbb{P}(M_\tau > H) \leq CH^{-p}(r + \log H)^p(\log H)^{p/2} + CH^{-M'}. \quad (40)$$

This dichotomy includes the terminal exit jump: when $\tau \leq N$, its endpoint occurs in that same free maximum. Integrate against $aH^{a-1}dH$, treating $H \leq 2r$ trivially and absorbing logarithms into an arbitrary small power of r . For all $0 < a < p$ and $\eta > 0$,

$$\mathbb{E}_{s,c} M_\tau^a \leq C_{a,\eta} r^{a+\eta}. \quad (41)$$

The jump tail is uniform conditionally on the past. Summing the events at times k with $\tau \geq k$, which are measurable before jump k , yields $\mathbb{P}(J_\tau > v) \leq C\mathbb{E}\tau e^{-cv}$ after a constant adjustment for the added one. By (39),

$$\mathbb{P}(J_\tau > v) \leq Cr^2 e^{-cv}, \quad \mathbb{E}J_\tau^a \leq C_a \log^a(r+2) \quad (a > 0). \quad (42)$$

The second bound integrates the minimum of the displayed tail and one.

6.3 The summable harmonic-defect potential

The exact exponent needed for the second-order correction is

$$\mathbb{E}_{s,c} \sum_{k < \tau} (1 + |S_k|)^{p-3} \leq C_\delta r^{p-1+\delta} \quad (\delta > 0). \quad (43)$$

Here are explicit uniform estimates. Set $q = p - 3 \in (1, 2)$ and $L_k = \log(k+2)$. For $k \leq r^2$, (37) bounds each expectation by Cr^q , so this part is $O(r^{p-1})$. For $k > r^2$ let $E_k = \{|S_k - s| > \sqrt{k}L_k\}$. Equations (38) with exponent 8 and (37) with moment $2q$, by Cauchy-Schwarz, give

$$\mathbb{E}[(1 + |S_k|)^q; E_k] \leq C(r + \sqrt{k})^q(k+2)^{-4} \leq Ck^{q/2-4} \leq Ck^{-3}. \quad (44)$$

Killing has been removed on this event; no conditional moment estimate is hidden. On E_k^c , $1 + |S_k| \leq C\sqrt{k}L_k$, so (36) with $M = 4$ gives

$$\mathbb{E}[(1 + |S_k|)^q; \tau > k, E_k^c] \leq Ck^{-3/2}L_k^q(r + L_k)^p + Ck^{q/2-4}L_k^q. \quad (45)$$

For every fixed $a \geq 0$,

$$\sum_{k > r^2} k^{-3/2} \log^a(k+2) \leq C_a r^{-1} \log^a(r+2).$$

This follows by integration, using $k = r^2 t$ and integrability of $t^{-3/2}(1 + \log t)^a$. The sum of (45) is therefore at most

$$C\{r^{p-1} \log^q(r+2) + r^{-1} \log^{p+q}(r+2) + 1\} \leq C_\delta r^{p-1+\delta}.$$

Together with (44), this proves (43) with all claimed uniformity.

7 Constructing the two killed harmonic functions

7.1 Second-order color corrections

The corrected one-step covariances Γ_P, Γ_Q are unequal. Their average is Σ , but pointwise isotropy does not follow from centering. Introduce the explicit mean-zero matrix correctors

$$B_P = \frac{1}{144} \begin{pmatrix} -7 & 55 \\ 55 & -115 \end{pmatrix}, \quad B_Q = \frac{1}{72} \begin{pmatrix} 7 & -55 \\ -55 & 115 \end{pmatrix}. \quad (46)$$

They satisfy

$$\sum_c \pi_c B_c = 0, \quad (I - P_0)B = \Gamma - \Sigma. \quad (47)$$

For the dual the corresponding matrices are

$$B_P^* = \frac{1}{9} \begin{pmatrix} 8 & -5 \\ -5 & -10 \end{pmatrix}, \quad B_Q^* = \frac{1}{9} \begin{pmatrix} -16 & 10 \\ 10 & 20 \end{pmatrix}, \quad (48)$$

and satisfy the same equations with Γ^* . All these identities follow by rational arithmetic from (17). On the color-mean-zero subspace $I - P_0$ acts as multiplication by $3/4$, which also explains the factor $4/3$ in the Poisson solutions.

7.2 Smooth extension and Taylor cancellation

Continue the analytic expression for U across both boundary rays in an angular neighborhood of the quadrant. Extend its smooth angular factor elsewhere around the circle, retaining its degree- p homogeneous radial factor, and define it as zero at the origin. Because $p > 4$, this is C^3 at the origin. Denote the extension again by U ; it agrees with (4) on the entire quadrant and vanishes on both boundary rays. For $j = 0, 1, 2, 3$,

$$|D^j U(s)| \leq C(1 + |s|)^{p-j}. \quad (49)$$

The analytic continuation is Σ -harmonic in that angular neighborhood away from the origin. A bounded color translation of any large quadrant point stays in this neighborhood.

On the full free state space define

$$\Lambda(s, c) = U(s + h_c) + \frac{1}{2} B_c : D^2 U(s + h_c), \quad f(s, c) = \mathbb{E}_{s,c} \Lambda(S_1, C_1) - \Lambda(s, c). \quad (50)$$

Then, uniformly on the quadrant,

$$|\Lambda(s, c) - U(s)| \leq C(1 + |s|)^{p-1}, \quad |f(s, c)| \leq C(1 + |s|)^{p-3}. \quad (51)$$

To check the cancellation, put $y = s + h_c$ and $\Delta Y = \Delta S + h_d - h_c$. The first-order Taylor term of U vanishes since $\mathbb{E}_c \Delta Y = 0$. The second-order terms, including the zeroth-order part from the matrix corrector, are

$$\frac{1}{2} \{ \Gamma_c + (P_0 B)_c - B_c \} : D^2 U(y) = \frac{1}{2} \Sigma : D^2 U(y) = 0$$

outside a fixed ball. The fixed ball contributes only $O(1)$. The third-derivative Taylor remainder is bounded by $C(1 + |s|)^{p-3}$ using (49) and the uniform exponential jump moments; along long Taylor segments one may use $(1 + |s| + |\Delta Y|)^{p-3}$ and integrate all jump powers. Variation of $D^2 U$ in the B term requires only $D^3 U$ and gives the same order. Both terms are evaluated at the corrected spatial argument; mixing the original and corrected increments would give a different, incorrect second-order calculation.

7.3 Exit integrability and stopped Dynkin identity

At exit, the position S_τ is within the last jump length of a point on the quadrant boundary. On that boundary $U = 0$. The gradient estimate, the bounded offsets, and the bounded matrices give

$$|\Lambda(S_\tau, C_\tau)| \leq C M_\tau^{p-1} J_\tau. \quad (52)$$

One may use the crossing point of the last segment as the boundary point. All points of the comparison segment have radius at most a fixed multiple of M_τ , so the angular extension causes no extra uncontrolled degree- p term.

For any prescribed $\delta > 0$, take

$$q_0 = 1 + \frac{1}{2(p-1)}, \quad q'_0 = 2p - 1, \quad (p-1)q_0 = p - \frac{1}{2} < p.$$

Apply (41) with $a = p - 1/2$ and $\eta = \delta q_0/2$, and (42) with $a = q'_0$. Hölder's inequality gives

$$\mathbb{E}[M_\tau^{p-1} J_\tau] \leq (\mathbb{E} M_\tau^{p-1/2})^{1/q_0} (\mathbb{E} J_\tau^{2p-1})^{1/(2p-1)} \leq C_\delta r^{p-1+\delta}. \quad (53)$$

Also (43) and (51) make the full defect sum absolutely integrable. Stopped Dynkin's identity is consequently legitimate and reads

$$\begin{aligned} \mathbb{E}_{s,c}[\Lambda(S_n, C_n); \tau > n] &= \Lambda(s, c) - \mathbb{E}_{s,c}[\Lambda(S_\tau, C_\tau); \tau \leq n] \\ &\quad + \sum_{k=0}^{n-1} \mathbb{E}_{s,c}[f(S_k, C_k); \tau > k]. \end{aligned} \quad (54)$$

Both terms on the right have limits by the integrability just proved.

For a fixed start, the difference between the left side and $\mathbb{E}[U(S_n); \tau > n]$ tends to zero. Indeed, on $|S_n - s| \leq \sqrt{n} \log(n+2)$, its absolute expectation is at most

$$C_s n^{-1/2} \log^{2p-1}(n+2) + o(1)$$

by (36) and (51). On the complementary event use (38) with exponent 12, the moment $2(p-1)$ in (37), and Cauchy–Schwarz; this is $O_s(n^{(p-1)/2-6}) = o(1)$. Thus the first limit in (5) exists and has the exact representation

$$V(s, c) = \Lambda(s, c) - \mathbb{E}_{s,c} \Lambda(S_\tau, C_\tau) + \mathbb{E}_{s,c} \sum_{k < \tau} f(S_k, C_k). \quad (55)$$

It is nonnegative because U is nonnegative on the quadrant. Equations (43), (51), and (53) prove (6), and choosing $\delta < 1$ gives $0 \leq V(s, c) \leq C(1 + |s|)^p$. The identical construction with h^*, B^* proves every assertion for V^* , including its representation and growth bound.

7.4 Harmonicity, normalization, and positive seed values

The stopped estimates also imply, uniformly in a starting state,

$$\sup_n |\mathbb{E}_{s,c}[U(S_n); \tau > n]| \leq C(1 + |s|)^p.$$

For example use (54) and the exit and potential bounds; the expectation of $|\Lambda - U|$ before exit is bounded by a moment of M_τ strictly below p . This polynomial majorant is integrable under one exponential-tailed jump. The one-step Markov property and dominated convergence therefore give

$$V(s, c) = \sum_{t \in Q, d} q_{cd}(t - s) V(t, d), \quad (56)$$

and likewise for V^* . Nonnegative harmonicity permits bounded-time optional stopping for the killed chain; the necessary integrability also follows from the polynomial growth and free moments.

The normalization is unique in the stated class. If another killed-harmonic function H satisfies $H - U = O((1 + |s|)^{p-1+\delta})$ for some $\delta < 1$, its killed expectation equals $H(s, c)$ at every finite time. On the event $|S_n - s| \leq \sqrt{n} \log(n + 2)$, (36) makes the expected difference from U tend to zero, since its polynomial degree is less than p . On the complement use (38) with exponent 16 and (37). Hence $H = V$. The same argument applies in reversal. This also shows that the choice of smooth extension outside the quadrant does not affect either normalized limit.

In any fixed interior angular sector, $U(s) \asymp |s|^p$, so (6) makes V positive sufficiently far out. From o , H steps $A_0 = (1, 0)$ followed by H steps $A_1 = (0, 1)$ reach (H, H, P) , stay in the quadrant, and have positive probability. By harmonicity, $V(o) > 0$. For the dual use $e = (1, 0, P)$ and the seed

$$(1, 0, P) \rightarrow (1, 0, Q) \rightarrow (0, 2, P) \rightarrow (2, 0, Q) \rightarrow (1, 3, P). \quad (57)$$

These four steps reverse C_1, D_2, C_3, D_3 , respectively. Thereafter the dual pair $(2, -2) : P \rightarrow Q$, $(-1, 3) : Q \rightarrow P$ has net displacement $(1, 1)$ and stays in the quadrant at every repetition. It reaches an arbitrarily deep fixed interior sector. Thus $V^*(e) > 0$ as well.

[write] **5 October 2026.** These are also Report 123’s seeds (Part II, Section 21). In the language of Theorem 24.1 of `a377922-corner-polyhedra-schnyder` they are seed paths in the sense of its (H5) from o for the forward walk and from e for the dual (Corollary 26.3); from the origin no dual seed exists (Remark 26.1).

8 Sharp survival and the endpoint meander

8.1 Deep entrance with quantitative slack

For a horizon n put, with harmless integer rounding,

$$h = n^{1/3}, \quad T = \lfloor n^{3/4} \rfloor, \quad R = n^{5/12}, \quad \nu = \inf\{k \geq 0 : S_k \in [h, \infty)^2\}. \quad (58)$$

Uniformly over states outside this deep set, the free process in time $\lfloor h^2 \rfloor$ has a probability bounded below of displacement less than $-2h$ in whichever coordinate is initially below h . This follows from (28), a nondegenerate Brownian marginal, and the finite number of colors. It is independent of the magnitude of the other coordinate by translation invariance. On this event the chain cannot both stay alive and remain outside the deep set for the whole block. Iterating the conditional bound over blocks gives

$$\mathbb{P}_{s,c}(\tau > T, \nu > T) \leq Ce^{-cT/h^2} \leq Ce^{-cn^{1/12}}. \quad (59)$$

No bounded-jump or ball-confinement assumption is made here.

For starts $|s| \leq n^{1/4}$, Brownian maximal tails up to T and the full exponential coupling inequality give

$$\mathbb{P}_{s,c} \left(\max_{k \leq T} |S_k| > R \right) \leq Ce^{-cn^{1/12}}. \quad (60)$$

Indeed $R^2/T = n^{1/12}$, $|s| = o(R)$, and the coupling excess tail at $R/4$ is still smaller. By Cauchy-Schwarz and the $2p$ free moment, the expectation of $(1 + \max_{k \leq T} |S_k|)^p$ on this event is at most $Cn^{3p/8}e^{-c'n^{1/12}}$, after reducing the positive constant c' . The identical weighted bound on $\{\tau > T, \nu > T\}$ follows from (59). Both are smaller than every prescribed negative power of n , uniformly in the full starting range.

Let

$$E_n = \{\nu \leq T, \tau > \nu, |S_\nu| \leq R\}.$$

By optional stopping at $\nu \wedge T$ and nonnegative harmonicity,

$$\mathbb{E}[V(S_\nu, C_\nu); \nu \leq T, \tau > \nu] = V(s, c) - \mathbb{E}[V(S_T, C_T); \tau > T, \nu > T]. \quad (61)$$

The last term and the large-radius contribution are superpolynomially small by the preceding weighted estimates. On E_n , $\min(S_{\nu,1}, S_{\nu,2}) \geq h$ and $|S_\nu| \leq R = o(\sqrt{n})$. The elementary wedge bound, transformed to physical coordinates, is

$$U(x) \geq c \operatorname{dist}(x, \partial \mathbb{R}_{\geq 0}^2) |x|^{p-1}.$$

It follows from $\sin(p\phi) \asymp \min(\phi, \theta - \phi)$ and norm equivalence. Choose $\delta = 1/4$ in (6). Then

$$\sup_{\substack{x \in [h, \infty)^2 \cap \mathcal{Q} \\ |x| \leq R, c=P, Q}} \left| \frac{V(x, c)}{U(x)} - 1 \right| \leq C \frac{R^{1/4}}{h} = O(n^{-11/48}) = o(1). \quad (62)$$

Thus, for every fixed start,

$$\mathbb{E}_{s,c}[U(S_\nu); E_n] \longrightarrow V(s, c). \quad (63)$$

Uniformly for $|s| \leq n^{1/4}$, this entrance expectation is at most $C(1 + |s|)^p$, since $U \leq 2V$ on the good set for all large n and (61) bounds its V -mass.

8.2 Fresh continuation coupling and the exact limit

Restart a free coupling from a good entrance state x over the remaining $n - \nu$ steps. Take error $e = D_M \log n + 1$. The two shifted Brownian starts $x \pm e(1, 1)$ remain inside the physical quadrant, and

$$\frac{U(x \pm e(1, 1))}{U(x)} = 1 + O(e/h) = 1 + o(1)$$

uniformly over the good set, by the gradient bound and the preceding boundary-distance lower bound. Their radii are $o(\sqrt{n})$, and $n - \nu = n(1 + o(1))$. The upper and lower shifted-path comparisons and the angle-uniform Brownian limit (31) therefore give

$$\mathbb{P}_{x,c}(\tau > n - \nu) = k_\theta U(x) n^{-p/2} (1 + o(1)) \quad (64)$$

uniformly over good positions, colors, and entrance times. The smallest principal mass is at least $ch^p n^{-p/2} = cn^{-p/6}$, so the coupling failure $O(n^{-M})$ is relatively negligible by taking $M > p/6 + 1$.

For a bounded Lipschitz endpoint test $f(AS_{n-\nu}/\sqrt{n})$ the same joint formula holds with k_θ replaced by $k_\theta \int f \varrho$. The coupled endpoint arguments differ by $O(\log n/\sqrt{n})$. Disagreement of survival indicators is bounded by the difference of the upper and lower Brownian survival probabilities plus the coupling-failure mass, which is $o(U(x)n^{-p/2})$ uniformly. This observation justifies signed tests too; a signed integral is not being compared solely by ordering its survival events. The time rescaling $\sqrt{(n - \nu)/n} \rightarrow 1$ is uniform and harmless under the L^1 limit.

Apply the Markov property at ν . Late and large entrance events cost $o(n^{-p/2})$ by (59)–(60). The uniform relative error in the continuation multiplies the bounded expected entrance U -mass. From (63), for every fixed start and every bounded Lipschitz f ,

$$n^{p/2} \mathbb{E}_{s,c}[f(AS_n/\sqrt{n}); \tau > n] \longrightarrow k_\theta V(s, c) \int_{\mathcal{W}} f(w) \varrho(w) dw. \quad (65)$$

In particular

$$n^{p/2} \mathbb{P}_{s,c}(\tau > n) \longrightarrow k_\theta V(s, c). \quad (66)$$

For $V(s, c) > 0$ this gives the corresponding conditional weak limit with density ϱ . If $V = 0$, the unnormalized mass is $o(n^{-p/2})$; no conditional positive-mass assertion is needed. Both statements hold identically for V^* and the reversed walk.

8.3 A bound uniform in every starting state

The same entrance argument gives, uniformly for $|s| \leq n^{1/4}$,

$$\mathbb{P}_{s,c}(\tau > n) \leq C(1 + |s|)^p n^{-p/2}.$$

The superpolynomial absolute errors can be absorbed since $1 + |s| \geq 1$. For $|s| > n^{1/4}$ the trivial bound one is at most $(1 + |s|)^{2p} n^{-p/2}$. Enlarging a constant for bounded n proves

$$\mathbb{P}_{s,c}(\tau > n) \leq C(1 + |s|)^{2p} n^{-p/2} \quad (s \in \mathcal{Q}, c = P, Q, n \geq 1). \quad (67)$$

The constant is uniform in both time directions. This bound is the specific endpoint-uniform input needed for the final infinite sum.

9 Free local bounds and last-window smoothing

9.1 Fourier inversion and removal of lattice periods

[write] **5 October 2026.** This subsection is also Report 123's Section 4.2 (Part II, Section 18.2).

Let $L(u)_{cd} = \sum_v q_{cd}(v) e^{iu \cdot v}$ for $u \in [-\pi, \pi]^2$. Exponential moments make L analytic near the real torus. Its spectral radius is at most one. If it has a unit-modulus eigenvalue ζ , equality in the triangle inequality, together with irreducibility, forces all components of its eigenvector to have the same modulus and, on every edge,

$$e^{iu \cdot v} f_d = \zeta f_c.$$

The $P \rightarrow P$ steps $A_0 = (1, 0)$ and $A_1 = (0, 1)$ give $e^{iu_1} = e^{iu_2} = \zeta$. The $Q \rightarrow Q$ step $D_1 = (1, -1)$ gives $\zeta = e^{i(u_1 - u_2)} = 1$. Hence $u = 0$ modulo 2π is the only peripheral point. This proves spatial and temporal aperiodicity directly.

Near zero the simple eigenvalue and its projection satisfy

$$\lambda(u) = 1 - \frac{1}{2} u^\top \Sigma u + O(|u|^3), \quad \Pi(u) = \mathbf{1}\pi + O(|u|).$$

The first expansion follows either from the corrector variance or from the log-Perron Hessian. The other eigenvalue is near $1/4$. Choose a small neighborhood of zero. Inside it the leading powers are bounded by $Ce^{-cm|u|^2}$ and the other part decays geometrically. Outside it compactness and the absence of peripheral points imply $\|L(u)^m\| \leq Cr^m$ for some $r < 1$; a slightly larger r absorbs any Jordan factors. Fourier inversion, $u = w/\sqrt{m}$, and dominated convergence now give

$$\mathbb{P}_c(S_m - S_0 = t, C_m = d) = \frac{\pi_d}{2\pi m \sqrt{\det \Sigma}} e^{-t^\top \Sigma^{-1} t / (2m)} + o(m^{-1}). \quad (68)$$

The error is uniform in every lattice t and both colors: the absolute Fourier-integral error does not depend on t . In particular all free endpoint probabilities are at most C/m , for $m \geq 1$ after enlarging C . Formula (23) gives the same LLT and bound for the dual.

9.2 A uniform off-diagonal free bound

We also need, for the free kernel p_k and all displacements,

$$p_k((x, c), (y, d)) \leq \frac{C}{k} \exp \left\{ -c \min \left(\frac{|y - x|^2}{k}, |y - x| \right) \right\} \quad (k \geq 1). \quad (69)$$

Here is a uniform small-real-tilt proof. Let $\mathcal{L}(z)_{cd} = \sum_v q_{cd}(v) e^{z \cdot v}$, let ϱ_λ be its positive Perron root at a small real vector λ , and put $\psi(\lambda) = \log \varrho_\lambda$. Analyticity follows from exponential tails, and

$$\psi(0) = 0, \quad \nabla \psi(0) = 0, \quad D^2 \psi(0) = \Sigma > 0.$$

Hence, in a small closed real ball, $\psi(\lambda) \leq C_0 |\lambda|^2$ and $D^2 \psi(\lambda) \geq c_0 I$.

If r_λ is a positive Perron right eigenvector, the tilted stochastic kernel

$$q_\lambda(c, d, v) = q_{cd}(v) e^{\lambda \cdot v} \frac{r_\lambda(d)}{\varrho_\lambda r_\lambda(c)}$$

has the same support as q . The self-loop phases A_0, A_1, D_1 therefore exclude all nonzero peripheral Fourier frequencies uniformly under these small real tilts, by the preceding edge-phase argument. Near zero analytic perturbation gives

$$|[\mathcal{L}(\lambda + it)^k]_{cd}| \leq C \varrho_\lambda^k (e^{-ck|t|^2} + \eta^k), \quad \eta < 1.$$

The linear imaginary term changes only the phase. Away from zero, compactness gives a uniform spectral gap after division by ϱ_λ . A resolvent contour exterior to that compact spectral set bounds matrix powers uniformly, including possible non-Perron eigenvalue collisions. Fourier inversion thus gives

$$p_k((x, c), (y, d)) \leq \frac{C}{k} e^{-\lambda \cdot (y-x) + k\psi(\lambda)}$$

for every small real λ ; Perron-vector ratios are bounded and absorbed into C . Put $\Delta = y - x$, $R = |\Delta|$ and $\lambda = \delta \Delta / \max(k, R)$ with small fixed $\delta > 0$ satisfying $C_0 \delta \leq 1/2$. For $R \leq k$ the exponent is at most $-(\delta/2)R^2/k$; for $R \geq k$ it is at most $-(\delta/2)R$. Take $\lambda = 0$ if $R = 0$. This proves (69) uniformly in colors. Reversal has the same bound, by its transposed Fourier matrix or the same argument.

9.3 A compact-interior killed local limit

Fix a starting state $x = (s, c)$ and a compact subset B of the open physical quadrant. Uniformly for $z/\sqrt{n} \in B$ and terminal colors d , we claim

$$K_n(x, (z, d)) = \frac{b_\theta \pi_d}{\sqrt{\det \Sigma}} V(x) n^{-p/2-1} u(Az/\sqrt{n}) e^{-|Az|^2/(2n)} + o(n^{-p/2-1}). \quad (70)$$

Fix $0 < \varepsilon < 1/2$, let $l = \lfloor \varepsilon n \rfloor$, $m = n - l$, and replace killing in the final window by the free kernel:

$$F_n(z, d) = \sum_{y \in \mathcal{Q}, a} K_m(x, (y, a)) p_l((y, a), (z, d)). \quad (71)$$

The uniform error in the free local limit (68), summed against K_m , costs only $o(l^{-1}) \mathbb{P}_x(\tau > m)$. For each fixed ε its contribution after multiplying by $n^{p/2+1}$ tends to zero. There is no unweighted summation of infinitely many local-limit errors.

Put $v = Az/\sqrt{n}$, $t = 1 - \varepsilon$, and $\phi_\varepsilon(v) = (2\pi\varepsilon)^{-1} e^{-|v|^2/(2\varepsilon)}$. Equations (65) and (68) give

$$\begin{aligned} n^{p/2+1} F_n(z, d) &\longrightarrow \frac{\pi_d k_\theta V(x)}{\sqrt{\det \Sigma}} H_\varepsilon(v), \\ H_\varepsilon(v) &= t^{-p/2} \int_{\mathcal{W}} \varrho(w) \phi_\varepsilon(v - \sqrt{t} w) dw. \end{aligned} \quad (72)$$

For fixed ε , the Gaussian test functions are bounded and Lipschitz in w , and uniformly Lipschitz in v on a compact target set. A finite-net argument makes convergence uniform there. The free leading coefficient is independent of its initial color a , so no additional conditioned-color mixing theorem is needed.

Extend ϱ by zero outside $\mathcal{W}$. This extension is bounded, continuous at the sides and apex, and tends to zero at infinity. Changing variables $y = \sqrt{t} w$ shows that H_ε is the convolution of ϕ_ε with $t^{-p/2-1} \varrho(y/\sqrt{t})$. Consequently $H_\varepsilon \rightarrow \varrho$ uniformly on compact sets as $\varepsilon \downarrow 0$. Since $k_\theta \varrho = b_\theta u e^{-|\cdot|^2/2}$, this is the profile in (70). If $V(x) = 0$, the free C/l bound instead shows directly that $n^{p/2+1} \sup_{z,d} F_n(z, d) \rightarrow 0$ for every fixed ε , using the zero-mass version of (66).

9.4 Controlling exit and re-entry during the last window

Choose $\eta > 0$ such that $\text{dist}(z, \partial\mathbb{R}_{\geq 0}^2) \geq \eta\sqrt{n}$ for $z/\sqrt{n} \in B$. Any first exit position w is outside the quadrant, so $|z - w| \geq \eta\sqrt{n}$, even for an unbounded overshoot. From (69), for ε sufficiently small relative to η^2 and then all sufficiently large n ,

$$\sup_{\substack{1 \leq k \leq l, \\ a=P, Q}} p_k((w, a), (z, d)) \leq \frac{C}{\varepsilon n} e^{-c\eta^2/\varepsilon}. \quad (73)$$

To verify it, set $R = |z - w|$. If $R \leq k$, maximize $k^{-1}e^{-c\eta^2 n/k}$ over $k \leq \varepsilon n$; for small enough ε it is increasing on this range and its maximum is the displayed bound. If $R > k$, use $k^{-1} \leq 1$ and $e^{-cR} \leq e^{-c\eta\sqrt{n}}$, smaller than that bound for fixed ε, η and large n . The case $k = 0$ contributes zero because w is outside and z is inside.

By strong Markov at the first exit,

$$F_n(z, d) - K_n(x, (z, d)) = \mathbb{E}_x[\mathbf{1}_{\{m < \tau \leq n\}} p_{n-\tau}((S_\tau, C_\tau), (z, d))].$$

Consequently

$$0 \leq F_n(z, d) - K_n(x, (z, d)) \leq \mathbb{P}_x(\tau > m) \frac{C}{\varepsilon n} e^{-c\eta^2/\varepsilon}. \quad (74)$$

After scaling by $n^{p/2+1}$, its limsup is at most $C_x \varepsilon^{-1} e^{-c\eta^2/\varepsilon}$ by (66). It tends to zero as $\varepsilon \downarrow 0$. Taking first $n \rightarrow \infty$ at fixed ε , then $\varepsilon \downarrow 0$, proves (70) with its stated compact-interior uniformity. All arguments apply equally to the reversed process.

Finally, split a path into two comparable pieces, remove killing in the latter half, and apply the free C/n bound. Fixed-start survival then gives the global endpoint bound

$$\sup_{z, d} K_n(x, (z, d)) \leq C_x n^{-p/2-1}. \quad (75)$$

Its constant may depend on the fixed start, which is sufficient below.

10 Fixed endpoints, domination, and the leading constant

10.1 Midpoint reversal and all normalization factors

For $n = 2m$, Chapman–Kolmogorov and (24) give the exact identity

$$K_{2m}((s, c), (t, d)) = \pi_d \sum_{z \in \mathcal{Q}, a} \frac{K_m((s, c), (z, a)) K_m^*((t, d), (z, a))}{\pi_a}. \quad (76)$$

On a compact interior midpoint set at scale $\sqrt{m}$, insert the two bulk limits (70). The whitened lattice has density $m\sqrt{\det \Sigma}$, since the cell of $Az/\sqrt{m}$ has area $1/(m\sqrt{\det \Sigma})$. Thus

$$\sum_z f(Az/\sqrt{m}) \sim m\sqrt{\det \Sigma} \int_{\mathcal{W}} f(w) dw.$$

The two factors contribute π_a^2 , division in (76) leaves π_a , and summing colors gives one. Using (35), the resulting coefficient is

$$\frac{\pi_d b_\theta^2}{\sqrt{\det \Sigma}} V(s, c) V^*(t, d) m^{-p-1} I_2 = \frac{\pi_d b_\theta}{\sqrt{\det \Sigma}} V(s, c) V^*(t, d) (2m)^{-p-1}.$$

There is neither an extra color multiplicity nor an omitted lattice Jacobian.

To justify removing the compact restriction, choose compact annular sectors $E_j \subset \mathcal{W}$ increasing to $\mathcal{W}$, with boundaries of zero limiting probability; for example restrict radius above and below and angle away from both sides. On the omitted region use (75) for the forward factor, and sum the dual factor. After multiplication by m^{p+1} its contribution is at most

$$C_{s,c} m^{p/2} \mathbb{P}_{t,d}^*(\tau^* > m, AS_m/\sqrt{m} \notin E_j).$$

If $V^*(t, d) > 0$, (65) bounds its limit by a constant times the ϱ -mass outside E_j , tending to zero. If $V^*(t, d) = 0$, the entire expression tends to zero by (66). This handles both boundary layers and large radii without a boundary-uniform killed local limit. A zero forward harmonic value is allowed too: its compact bulk coefficient is zero, and the same truncation controls the complement.

For odd n , use times $m = \lfloor n/2 \rfloor$ and $j = \lceil n/2 \rceil$. On a common compact midpoint set $j/m \rightarrow 1$, and homogeneity and the continuous Gaussian factors give the same limiting product. The lattice density is still $m\sqrt{\det \Sigma}$ and $n/m \rightarrow 2$. The complementary bound is unchanged. Hence (7) holds along all integers, with no parity factor.

10.2 A polynomial bound uniform in the terminal point

Write $n = i + j + k$ with the three positive parts comparable to n . Remove killing in the middle part and use its free C/j bound. Reversing the last part gives

$$\sum_{v,b} K_k((v, b), (t, d)) = \pi_d \sum_{v,b} \frac{K_k^*((t, d), (v, b))}{\pi_b} \leq \frac{\pi_d}{\min_b \pi_b} \mathbb{P}_{t,d}^*(\tau^* > k).$$

Therefore

$$K_n(o, (t, d)) \leq \frac{C}{n} \mathbb{P}_o(\tau > i) \mathbb{P}_{t,d}^*(\tau^* > k). \quad (77)$$

Use fixed-start survival (66) for o and the global estimate (67) for (t, d) . This proves (8) for all large n uniformly in t, d ; increasing C covers the finitely many smaller n .

10.3 Summing the exact endpoint weights

For $m = n - 1$, the exact identity (18) is

$$a_n = 2 \cdot 9^{n-1} \sum_{t,d} g(t, d) K_m(o, (t, d)).$$

The endpoint bound dominates $m^{p+1} g(t, d) K_m(o, (t, d))$ by $C g(t, d) (1 + |t|)^{2p}$, a summable function on the quadrant and colors. Dominated convergence using (7) now gives

$$\lim_{m \rightarrow \infty} m^{p+1} \sum_{t,d} g(t, d) K_m(o, (t, d)) = \frac{b_\theta}{\sqrt{\det \Sigma}} V(o) \sum_{t,d} \pi_d g(t, d) V^*(t, d).$$

This proves (9)–(10). The original length shift supplies exactly $2/9$, since $(n/(n-1))^{p+1} \rightarrow 1$. Finiteness also follows directly from the polynomial growth of V^* and the exponential decay of g . Strict positivity uses $V(o) > 0$ and the single positive summand $e = (1, 0, P)$, where $g(e) = 1/6$, $\pi_P = 2/3$, and $V^*(e) > 0$. All assertions of Theorem 1.1 are proved.

For clarity, the same constant can be written

$$C_A = \frac{2^{2-p}\sqrt{3}}{45\theta\Gamma(p+1)} V(o) \sum_{t,d} \pi_d g(t,d) V^*(t,d).$$

This algebraic form introduces no additional normalization. It is not an effective numerical evaluation: the proof establishes existence, convergence, and positivity without an explicit truncation error for computing the harmonic values.

11 The vanishing-error inverse ceiling bracket

Appending the new largest legal entry n to a length- n avoider is injective and cannot create a triple satisfying (1). Thus $a_n \leq a_{n+1}$. The leading equivalent makes it unbounded, so $N(y)$ is finite for all large y .

Put $\lambda = \log 9$, $L = \log y$, and

$$f(x) = \lambda x - \kappa \log x + \log C_A, \quad \eta_n = \log a_n - f(n) \quad (n \geq 1).$$

Equation (9) is exactly $\eta_n \rightarrow 0$. Let $b(L) = \kappa \log L - \kappa \log \lambda - \log C_A$, so that $v = (L + b(L))/\lambda$ is (11). Its deterministic centering defect is

$$f(v) - L = -\kappa \log(1 + b(L)/L) = O(\log L/L) = o(1). \quad (78)$$

For large L define

$$\begin{aligned} m(L) &= \lfloor L/(2\lambda) \rfloor, & E(L) &= \sup_{n \geq m(L)} |\eta_n|, \\ D(L) &= E(L) + |f(v) - L|, & \varepsilon(y) &= 4D(L)/\lambda + 1/L. \end{aligned} \quad (79)$$

This is a well-defined positive envelope once $m(L) \geq 1$. Since $\eta_n \rightarrow 0$, it satisfies $E(L) \rightarrow 0$, $D(L) \rightarrow 0$, and $\varepsilon(y) \rightarrow 0$. It is an existence envelope, without a claimed computable rate or finite onset.

For all sufficiently large L , $v - \varepsilon$ and $v + \varepsilon$ exceed $m(L)$, and

$$f'(x) = \lambda - \kappa/x \geq \lambda/2 \quad (x \geq m(L)).$$

For every integer $m(L) \leq n < v - \varepsilon$,

$$\begin{aligned} \log a_n &\leq f(n) + E(L) \leq f(v - \varepsilon) + E(L) \\ &\leq f(v) - \lambda\varepsilon/2 + E(L) < L. \end{aligned}$$

The strict inequality follows from the definition of D and the positive $1/L$ margin. For the smaller indices, monotonicity and

$$\log a_{m(L)} = L/2 + O(\log L) + o(1) < L$$

give the same conclusion. Every integer $n < v - \varepsilon$ is therefore below the threshold, proving $N(y) \geq \lceil v - \varepsilon \rceil$. For $n_+ = \lceil v + \varepsilon \rceil$,

$$\begin{aligned} \log a_{n_+} &\geq f(n_+) - E(L) \geq f(v + \varepsilon) - E(L) \\ &\geq f(v) + \lambda\varepsilon/2 - E(L) > L. \end{aligned}$$

Hence $N(y) \leq n_+$ and Corollary 1.2 follows.

If $[v - \varepsilon, v + \varepsilon]$ contains no integer, both bounding ceilings coincide with $\lceil v \rceil$. Near an integer the unknown coefficient remainder can choose between neighboring indices, so the leading equivalent alone does not justify a universal equality $N(y) = \lceil v(y) \rceil$. The constant term in the real center is $(-\kappa \log \lambda - \log C_A)/\lambda$. The explicit $O(\log L/L)$ defect in (78) may be smaller than the unknown $\eta_n = o(1)$; it cannot be promoted to a justified next inverse correction without a quantitative coefficient remainder.

12 The unique singularity on the critical circle

Put $\rho = 1/9$. Besides the leading equivalent, the exact walk proves:

Corollary 12.1. *The function F has radius ρ , is singular at ρ , and admits analytic continuation through every other point on $|z| = \rho$. Moreover p, κ are irrational and*

$$\lim_{x \uparrow \rho} \frac{\log F^{(5)}(x)}{\log(1 - 9x)} = \kappa - 6. \quad (80)$$

The function F is not D -finite, even over $\mathbb{C}(z)$.

[write] **5 October 2026.** Sections 12 and 13 are Report 123's Sections 8 and 9, word for word except that Report 123 has no corollary here (these statements are part of its Theorem 15.1) and that it cites its logarithmic law (89) where this text cites (9): in Section 12, for the radius $1/9$ by Cauchy–Hadamard, and in Section 13, for the bounds $n^{5-\kappa \pm \varepsilon} t^n$ on the summands. Both uses need only $a_n = 9^n n^{-\kappa + o(1)}$, so Corollary 12.1 does not depend on the equivalent; it was proved first in Report 123, from its own logarithmic law.

12.1 Uniform return loops

Apart from o , every reachable shifted state lies in

$$\mathcal{D} = \{P(a, b) : a, b \geq 0, a + b \geq 1\} \cup \{Q(a, b) : a \geq 1, b \geq 0\}.$$

A Q target always has first coordinate at least one. There is no incoming edge to o from $\mathcal{D}$: an A_d target at $(0, 0)$ would require $d = b = 0$ and $a = -1$, while a C_d target there would require $d = 1, b = a = 0$ in color Q , which is absent. Let T be the killed transition operator restricted to $\ell^\infty(\mathcal{D})$. It is positive and $\|T\|_\infty \leq 1$.

Every state has a retained zero-displacement return loop of length three and one of length four, as shown by these edge sequences:

Starting state	Three steps	Four steps
$P(a, b), a \geq 1$	A_2, D_2, C_1	A_1, A_2, D_3, C_1
$P(0, b), b \geq 1$	A_1, D_2, C_2	A_1, A_1, D_3, C_2
$Q(a, b), a \geq 1$	C_1, A_2, D_2	C_1, A_1, A_2, D_3

For example, the first triple has displacements $(-1, 2), (1, -2), (0, 0)$; its intermediate states are legal for all $a \geq 1, b \geq 0$. For the second triple, the positions relative to $(0, b)$ are $(0, 1), (1, -1), (0, 0)$,

legal for $b \geq 1$. The other cases follow by the same explicit additions. All intermediate colors and positions remain in $\mathcal{D}$. Since the tilt telescopes around a closed loop, the probability of any listed length- l loop is exactly 9^{-l} . As positive operators,

$$T^3 \geq 9^{-3}I, \quad T^4 \geq 9^{-4}I. \quad (81)$$

12.2 The operator resolvent away from the positive point

For $l = 3$ or 4 , write $T^l = \epsilon_l I + B_l$, where $\epsilon_l = 9^{-l}$, $B_l \geq 0$, and $\|B_l\|_\infty \leq 1 - \epsilon_l$. If $|t_0| = 1$ and $t_0^l \neq 1$, then

$$\frac{|t_0|^l(1 - \epsilon_l)}{|1 - \epsilon_l t_0^l|} < 1.$$

The strict inequality persists in an open neighborhood of t_0 . There the norm-convergent series

$$(I - t^l T^l)^{-1} = \frac{1}{1 - \epsilon_l t^l} \sum_{j \geq 0} \left(\frac{t^l B_l}{1 - \epsilon_l t^l} \right)^j \quad (82)$$

is analytic. Multiplying by $I + tT + \dots + t^{l-1}T^{l-1}$ produces a two-sided analytic inverse of $I - tT$, since all factors are polynomials or norm limits of polynomials in T . The only unit-modulus number with both $t^3 = t^4 = 1$ is 1. Thus every other unit-circle point has such a resolvent neighborhood.

The first transition from o has just two retained targets; as a row functional on $\mathcal{D}$ its nonzero entries are

$$v(P(1, 0)) = 1/3, \quad v(P(0, 1)) = 5/27.$$

Separating a_0, a_1 in (18) gives, initially for $|z| < 1/9$,

$$F(z) = 1 + z + 18z^2 v(I - 9zT)^{-1}g. \quad (83)$$

Here $g|_{\mathcal{D}}$ is bounded and v has finite support, so the operator continuation is an ordinary scalar analytic continuation of F . Equation (9) gives radius exactly $1/9$ by Cauchy–Hadamard. Equation (82) excludes all its convergence-circle points except $1/9$. The next section establishes that the latter is genuinely singular.

This circle control does not claim a local Δ -domain expansion at $1/9$ and is not used to extract a leading amplitude.

13 Irrationality and the arithmetic obstruction

13.1 The irrational exponent

Since $\cos(\pi/4) < 2/\sqrt{7} < \cos(\pi/5)$, one has $\pi/5 < \theta < \pi/4$ and hence $4 < p < 5$, $5 < \kappa < 6$. The right-hand trigonometric comparison may be squared using $\cos(\pi/5) = (1 + \sqrt{5})/4$; both sides are positive. Moreover

$$\cos(2\theta) = 2\cos^2 \theta - 1 = 1/7.$$

If θ/π were rational, $e^{2i\theta}$ would be a root of unity, so $e^{2i\theta} + e^{-2i\theta} = 2/7$ would be an algebraic integer. A rational algebraic integer is an integer, a contradiction. Thus θ/π , its reciprocal p , and $\kappa = 1 + p$ are irrational.

13.2 A power bound for the fifth derivative

For $t \uparrow 1$,

$$F^{(5)}(t/9) = 9^5 t^{-5} \sum_{n \geq 5} (n)_5 a_n 9^{-n} t^n, \quad (n)_5 = n(n-1) \cdots (n-4).$$

All coefficients are positive. For any $0 < \varepsilon < 6 - \kappa$, (9) and $(n)_5 \asymp n^5$ bound the summands, up to fixed positive factors, by $n^{5-\kappa-\varepsilon} t^n$ and $n^{5-\kappa+\varepsilon} t^n$. For every fixed $\alpha > -1$,

$$\sum_{n \geq 1} n^\alpha t^n \asymp (1-t)^{-\alpha-1} \quad (t \uparrow 1). \quad (84)$$

To see this directly, put $s = -\log t \asymp 1 - t$. A lower bound follows from $1/s \leq n \leq 2/s$. An upper bound follows by splitting into unit intervals in ns , or comparing with the integrable function $x^\alpha e^{-x}$ after separately treating $ns \leq 1$. Finite omitted terms are absorbed. Therefore

$$c_\varepsilon (1-t)^{\kappa-6+\varepsilon} \leq F^{(5)}(t/9) \leq C_\varepsilon (1-t)^{\kappa-6-\varepsilon}. \quad (85)$$

The lower bound diverges. Taking logarithms, dividing by the negative number $\log(1-t)$, and then letting $\varepsilon \downarrow 0$ proves (80). In particular F cannot be analytic at $1/9$, completing the unique-dominant-singularity assertion.

13.3 The precise G function implication

Suppose first that F is D-finite over $\mathbb{Q}(z)$. Its coefficients are integers and exponentially bounded by (18); all their algebraic conjugates equal themselves and the common denominators are one. These facts, together with hypothetical D-finiteness, verify the G-function conditions. The same is true of $F^{(5)}$: its coefficients are integers, its polynomial coefficient factor preserves an exponential bound, and differentiation preserves D-finiteness.

We use the André–Chudnovski–Katz theorem in the precise formulation of Fischler–Rivoal [6, Theorem 6, pp. 328–329]. A minimal operator for a G-function over an algebraic extension of $\mathbb{Q}$ is Fuchsian with rational local exponents. At each finite point its solutions have finite sums of the form

$$\sum_{q \in E} \sum_{k=0}^{K_q} w^q (\log w)^k f_{qk}(w), \quad E \subset \mathbb{Q} \text{ finite}, \quad (86)$$

where the f_{qk} are holomorphic near 0 on the chosen local branch. The leading power-log conclusion is also stated there as Corollary 1. We need only the rationality of that leading power, not its arithmetic coefficient.

Apply this at $w = \rho - x \downarrow 0$ to $F^{(5)}$. To handle possible cancellation explicitly, first combine exponent classes modulo integers in (86), then expand their finitely many holomorphic coefficients. The union of their exponent supports is contained in a finite union of sets $q + \mathbb{N}$ and is bounded below. Since the function is nonzero, there is a smallest nonvanishing power $q_0 \in \mathbb{Q}$ and, at that power, a largest nonvanishing logarithmic degree $k_0 \geq 0$. Higher powers and lower logarithmic degrees are negligible along the positive real w axis. Thus

$$F^{(5)}(\rho - w) = c w^{q_0} (\log w)^{k_0} (1 + o(1)), \quad c \neq 0.$$

Its positivity permits taking its real logarithm; equivalently take absolute values first. Its logarithmic growth exponent must therefore be $q_0 \in \mathbb{Q}$. But (80) gives the irrational number $\kappa - 6$. This

contradiction proves non-D-finiteness over $\mathbb{Q}(z)$. Irrational powers of arbitrary complex D-finite functions would not supply this contradiction; the G-function hypotheses are the necessary arithmetic input.

13.4 Descent from complex to rational coefficients

If a rational-coefficient formal series satisfies a nonzero differential operator over $\mathbb{C}(z)$, clear its denominators and fix the finite order and polynomial-degree bounds of that operator. The unknown operator coefficients then form a finite vector over $\mathbb{C}$. Equating every formal series coefficient to zero gives homogeneous linear equations with rational coefficients, possibly infinitely many. Their rational row space is finite dimensional, so finitely many of those rows form a basis. Its rank is unchanged by extending scalars from $\mathbb{Q}$ to $\mathbb{C}$. A nonzero complex null vector therefore implies a nonzero rational null vector of the same finite system and thus of all its equations. This would give a nonzero operator over $\mathbb{Q}(z)$ annihilating F , contrary to what was just proved. Hence F is not D-finite over $\mathbb{C}(z)$ either.

14 Scope, reproducibility, and further questions

The principal theorem is the full coefficient equivalent (9), with a finite strictly positive constant characterized by (10). Its proof includes original-time survival and fixed-endpoint limits, not merely a logarithmic exponent. None of the limits above has been assigned an effective convergence rate. Therefore this report does not certify amplitude digits, an explicit numerical onset, a $1/n$ correction, a correction transseries, or a local complex Δ -domain expansion at the positive singularity. Circle continuation alone is not such a local expansion.

The argument suggests studying analogous finite-color cone processes, but no general Markov-additive cone theorem is asserted here. Extending the method requires separate checks of regeneration, harmonic regularity, corrector solvability, entrance estimates, lattice structure, and the endpoint weights in each proposed setting. These are further questions, not additional consequences claimed for an unaudited class of models.

[write] 5 October 2026. At the logarithmic scale a general theorem of this kind is in the repository: Theorem 24.1 of `a377922-corner-polyhedra-schnyder` (bundle Report 112), whose method is Part II's. Section 26 records how this walk meets its hypotheses and extends it to fixed endpoint states; Remark 26.4 lists where this Part uses $4 < p < 5$. No general amplitude theorem is claimed here either.

The companion package supplies the editable source, rendered PDF, exact integer data, and offline finite checks. The two kinds of evidence remain distinct. Rational identities and finite enumerations are exactly replayable; strong approximation, stopped integrability, angular uniformity, smoothing, and limiting domination are mathematical proofs in this text, not consequences of matching finitely many coefficients.

The precise finite ranges and fixture provenance are documented in `checks/README.md`. The checker tests the combinatorial model and exact probability algebra, including both first- and second-order color correctors, reversal, regenerative covariance, whitening, and the normalization and inverse identities. Its guards remain active under optimized Python. Independent negative tests mutate mathematical data and inventory conditions. Both the checks subtree and the complete package have closed bitwise inventories that reject extra files or directories, missing files, changed bytes, and symbolic links. Inventories are integrity records rather than cryptographic signatures.

The PDF builder runs twice in clean external directories, rejects overfull boxes and unresolved references, and requires byte equality. The complete replay runs exact checks in normal and optimized Python, compares the authoritative result bytes, rebuilds the PDF, creates a deterministic ZIP, extracts it into a fresh directory, reruns checks and rebuilds, and requires byte-identical repacking. The original package is verified again at the end. Commands, dependencies, and distinctions between full and checks-only replay appear in `README.md`. Downloaded source papers, intermediate render images, fitted diagnostics, and private development material are not reader-package contents.

[write] **5 October 2026.** In this repository the delivered `checks/README.md` is `125-equiv-checks-README.md` and the checker is `code/125-equiv-checks-verify.py`; the PDF, the inventories and the `checks/legacy/` copy of Report 123's checks are not shipped (Report 123's checks are shipped once, under the prefix `123-logexp-`), and the delivered `README.md` is replaced by this report's guide. The delivered scripts read the delivered layout, so they run only on a copy of the delivery; the report's `README` gives the routes.

14.1 Further questions and research

[write] **5 October 2026.** This subsection is added in the write. Under Vladimir's standing rule of 4 October 2026 it collects every claim of this Part that is stated without proof, and the questions this Part leaves open, with source, sketch and what is missing. The intake found no false claim in Report 125. Items common to both Parts are stated here once; Section 27 adds Part II's own.

1. **Digits of C_A .** Report 125 (Sections 10 and 14) characterizes C_A by (10) and says that the proof gives no truncation error for computing $V(o)$ or the series $\sum_{t,d} \pi_d g(t,d) V^*(t,d)$, so no digit is certified. Missing: an effective evaluation of the killed harmonic functions, for example a rate in the limits (5) or a computable representation (55) with explicit tails. Prior numerical estimates of exponent and amplitude are Kotěšovec's (OEIS, 13 January 2026, after Beaton) and Britt and Beaton's (their Section 4.2 and Appendix B.3), credited by both Parts; they are not reproduced or checked here. *Observation, not a claim* (intake, double precision on the exact b-file terms): $a_n 9^{-n} n^\kappa = 77.91, 83.24, 86.55, 87.65$ at $n = 100, 200, 500, 1000$, increasing; a two-point extrapolation in $1/n$ from $n = 500$ and 1000 gives about 88.76 , but the order of the correction is unknown, so this is not a digit of C_A .
2. **A remainder rate and corrections.** Is $a_n/(C_A 9^n n^{-\kappa}) - 1 = O(n^{-\delta})$ for some $\delta > 0$, and what is the first correction (a $1/n$ term or another scale), or a correction transseries? Report 125, Section 14, and Report 123, Section 15 ("What is not asserted"), leave this open. Missing: rates in every limit of the proof (strong approximation transfer, deep entrance, angular uniformity, last-window smoothing, dominated endpoint sum). *Observation, not a claim*: the secant exponent $-\log(a_n 9^{-n}/a_m 9^{-m})/\log(n/m)$ is 5.3834 at $(n,m) = (1000, 500)$, against $\kappa = 5.40169$.
3. **A local expansion at $1/9$.** Both Parts say that their circle control (Section 12) is not a local Δ -domain expansion. Is F continuable to a Δ -domain at $1/9$ with a singular expansion $c(1-9z)^{\kappa-1}(1+o(1))$ plus analytic terms? Missing: control of the resolvent (83) near $z = 1/9$ off the real axis; Report 123's kernel (Section 16.2) is a possible algebraic entry point.
4. **The ceiling rule.** Report 125 (Section 11) proves only the bracket (12) and says that $N(y) = \lceil v(y) \rceil$ holds whenever $[v - \varepsilon, v + \varepsilon]$ contains no integer, and that the $O(\log L/L)$ centring defect cannot be promoted to a next inverse correction. Is $N(y) = \lceil v(y) \rceil$ for all large y , or for a set of y of full logarithmic density? Missing: a quantitative remainder in item 2.

Observation, not a claim: with the uncertified value 88.76 in place of C_A , the intake found $N(y) = \lceil v(y) \rceil$ at $y = 10^{100}, 10^{300}, 10^{600}, 10^{900}$.

5. **Effective onsets.** The envelope $\varepsilon(y)$ of Corollary 1.2 is an existence envelope without a computable rate or onset (Section 11); Part II's n_ε (Theorem 15.1) and n_0 (Lemma 18.1) are not explicit either. Missing: explicit constants in the functional limit and coupling inputs.
6. **A general amplitude theorem.** Report 125, Section 14, suggests studying analogous finite-colour cone processes and lists what each would need: regeneration, harmonic regularity, solvability of the correctors, entrance estimates, lattice structure and the endpoint weights. This is Question Q1 of `a377922-corner-polyhedra-schnyder` (`cps:log:q:amplitude`). Missing beyond the checks named there: this Part uses $4 < p < 5$ and $p > 2$ (Remark 26.4); for other exponents the corrector Λ of (50) and the potential bound (43) need another construction.
7. **External inputs not re-checked at the write.** Bashtova and Shashkin's strong approximation (SPA 150 (2022), Theorem 2.1; arXiv:2006.09583v1, Conditions (A), (B), Theorem 1), including the statement that it needs no independence or nonlattice condition and controls the whole path; the normalization of Bañuelos and Smits' wedge kernel (Lemma 1, equation (2.2)); Whitt's Theorem 2.1(ii) (Part II); Britt and Beaton's equations (2.44)–(2.46), Section 4.2 and Appendix B.3 (arXiv:2512.21943v3; not retrieved at the write); the OEIS attributions. The locators of Fischler and Rivoal (Theorem 6 on p. 328 and Corollary 1 on p. 329 of the journal version; Theorem 3 and Corollary 1 in arXiv:1103.6022v2) were confirmed at the independent check of `a377922-corner-polyhedra-schnyder`. Two arguments of this Part are condensed and were read as sound at the intake, not expanded: the angular uniformity of Section 5 ((30)–(34)) and the tilted-resolvent bound of Section 9 ((69)).

Part II

The exact logarithmic exponent: a second route

Bundle Report 123, archive `A279571_Exact_Asymptotic_Exponent_and_Non_D_Finiteness_Source.zip`, delivered as `report123.tex` (16 pages), dated 2 October 2026. Delivered title “A279571 exact logarithmic exponent”, subtitle “Unique dominant singularity and non D finiteness”, with the header “REPORT 123” and the line “Standalone proof and exact finite check companion”. Printed with `[write]` notes. Report 123’s Section k is Section $k + 14$ here, and its Lemma $k.j$ is Lemma $(k + 14).j$; its equations continue the report’s numbering. Its Sections 2.1, 3.1, 3.2, 4.2, 8 and 9 are Part I’s text and are replaced by pointers. Sections 26 and 27 are added in the write.

Abstract of this Part (as delivered). Let a_n count inversion sequences of length n avoiding 100, 101, and 201. We prove the logarithmic asymptotic law

$$a_n = 9^n n^{-\kappa+o(1)}, \quad \kappa = 1 + \frac{\pi}{\arccos(2/\sqrt{7})}.$$

The generating function has its unique convergence-circle singularity at $1/9$ and is not D-finite, even over $\mathbb{C}(z)$. An exact positive change of weights turns the published two-color generating tree into a centered Markov-additive walk in a quadrant. Its corrected covariance gives an irrational wedge exponent. Free limit theorems, uniform annulus estimates, and two stopped arms joined at the original integer time prove matching logarithmic bounds. Uniform loops of lengths three and four control the rest of the critical circle. A G-function argument uses the irrational real growth exponent of the fifth derivative. We also prove the two-term logarithmic inverse. The argument does not establish a leading constant, a coefficient equivalent, or a correction expansion. Previously published numerical exponent and amplitude estimates are explicitly credited.

Reading conventions. As in Part I, except that W is a whitening matrix (any real W with $W\Sigma W^\top = I$; Part I’s A is one), $\rho = 1/9$, $\|s\|_W = |Ws|$, and J is an angular interval. This Part’s H, J, P, Q, K, B in Section 16.2 are generating functions and polynomials, not Part I’s objects (front matter, “Notation across the two Parts”).

15 Definitions and main theorem

An inversion sequence of length n is an integer sequence $(e_1, \dots, e_n)$ with $0 \leq e_i < i$. The forbidden patterns here are exactly the triples $i < j < k$ satisfying

$$e_i > e_j, \quad e_j \leq e_k, \quad e_i \geq e_k. \quad (87)$$

Indeed $e_j = e_k$, $e_i = e_k$, and the strict intermediate case give 100, 101, and 201, respectively. Let a_n count the avoiders, $a_0 = 1$, and $F(z) = \sum_{n \geq 0} a_n z^n$. This is OEIS A279571 and Class 2106 of Britt and Beaton [1, 2]. The initial terms are

$$1, 1, 2, 6, 22, 92, 424, 2106, 11102, 61436, 353980, 2110366, \dots$$

Throughout, logarithms are natural. Put

$$\theta = \arccos \frac{2}{\sqrt{7}}, \quad p = \frac{\pi}{\theta}, \quad \kappa = 1 + p, \quad \rho = \frac{1}{9}. \quad (88)$$

Theorem 15.1. *The numbers p and κ are irrational, with $4 < p < 5$. For every $\varepsilon > 0$ there are constants $c_\varepsilon, C_\varepsilon > 0$ and n_ε such that*

$$c_\varepsilon 9^n n^{-\kappa-\varepsilon} \leq a_n \leq C_\varepsilon 9^n n^{-\kappa+\varepsilon} \quad (n \geq n_\varepsilon). \quad (89)$$

Equivalently, $a_n = 9^n n^{-\kappa+o(1)}$. The function F has radius ρ , admits analytic continuation through every point of $|z| = \rho$ other than ρ , and is singular at ρ . Moreover,

$$\lim_{x \uparrow \rho} \frac{\log F^{(5)}(x)}{\log(1-9x)} = \kappa - 6, \quad (90)$$

and F is not D -finite over $\mathbb{C}(z)$. For $N(y) = \min\{n \geq 0 : a_n \geq y\}$,

$$N(y) = \frac{\log y}{\log 9} + \frac{\kappa}{\log 9} \log \log y + o(\log \log y) \quad (y \rightarrow \infty). \quad (91)$$

[write] **5 October 2026.** Part I supersedes every item of this theorem. The bounds (89) follow from (9). The circle statement, (90) and non- D -finiteness are Corollary 12.1, whose proof is this Part's own Sections 8–9, printed once in Part I (Sections 12–13). The inverse (91) follows from Corollary 1.2 with error $O(1)$. The proof below is the earlier and weaker route, the first proof of the exponent in the bundle. It is also an instance of the proof, though not of the statement, of Theorem 24.1 of `a377922-corner-polyhedra-schnyder` (Section 26).

What is not asserted. There is no proof here of $a_n \sim C 9^n n^{-\kappa}$, of the existence or value of C , or of any all-order correction or transseries law. The proved estimate allows bounded positive log-periodic modulation, logarithmic factors, and more generally $\exp(o(\log n))$ factors. It excludes a different power of n and a real factor $\exp(cn^\delta)$ with fixed $c \neq 0$, $\delta > 0$. No effective numerical threshold n_ε is provided. The uniform constants needed in the proof are quantified by their dependencies below.

[write] **5 October 2026.** This paragraph describes Report 123 alone. Part I proves $a_n \sim C_A 9^n n^{-\kappa}$ with $0 < C_A < \infty$ (Theorem 1.1); this excludes bounded log-periodic modulation, logarithmic factors and every factor $\exp(o(\log n))$ that does not tend to a positive constant. Part I also gives the inverse with error $O(1)$ and an explicit centre (Corollary 1.2). Still open in both Parts: the value of C_A , corrections, any transseries, and effective onsets such as n_ε (Section 14.1, items 1, 2 and 5).

Attribution. The exact combinatorial input is the generating tree in Britt–Beaton [2], version 3, 29 September 2026, Section 2.5, equation (2.44). Their Section 4.2 and Appendix B.3 give numerical asymptotic estimates. The OEIS formula credits Václav Kotěšovec, 13 January 2026, after Nicholas R. Beaton, for refined numerical exponent and amplitude estimates. Those prior estimates are not new numerical results of this report. We do not reproduce or refine amplitude digits, and numerical fitting plays no role in the proof.

16 From the generating tree to a quadrant walk

16.1 Exact combinatorial coordinates

[write] **5 October 2026.** Report 123's Section 2.1 is word for word Part I's Section 2.1, with the rules (13) and the steps (14); it is printed there.

16.2 Functional identities and the exact kernel

For this subsection only use the unshifted coordinates, including the empty root. Let $P(z, x, y)$ and $Q(z, x, y)$ mark size and those two coordinates; set

$$H(z, x) = P(z, x, 1) + Q(z, x, 1), \quad J(z, y) = yP(z, y, y) + Q(z, y, y).$$

For a monomial $x^a y^b$ of color P , the first rule contributes $\sum_{d=0}^a x^{a+1-d} y^{b+d}$, and the other two rules give analogous finite geometric sums. Thus, as coefficientwise formal identities,

$$P = 1 + \frac{zx}{x-y}(xP + Q - J), \quad Q = \frac{zx}{1-y}(H - P - Q). \quad (92)$$

The denominators are harmless after multiplication: each coefficient is a polynomial. Eliminating P and Q gives the kernel

$$\begin{aligned} K(z, x, y) &= (x-y)(1-y) - zx^2(1-y) + zx(x-y) - z^2x^2(x-1) \\ &= y^2 + (zx^2 - zx - x - 1)y + x - z^2x^2(x-1), \end{aligned} \quad (93)$$

and, writing $B = P + Q$,

$$KB = (1-y)(x-y-zxJ) + zx\{x-y-zx(x-1)\}H. \quad (94)$$

On a kernel branch where the evaluations and divisions are valid,

$$J = A(z, x, y)H + B_0(z, x, y), \quad A = \frac{x-y-zx(x-1)}{1-y}, \quad B_0 = \frac{x-y}{zx}.$$

Two x roots at fixed y therefore give the affine relation $H(x_j) = (A_i H(x_i) + B_{0,i} - B_{0,j})/A_j$ when $A_j \neq 0$. At exceptional points the cleared identity (94) is the appropriate statement. No global contraction property is assumed.

Direct expansion gives

$$\text{Disc}_y K = (x-1)(zx-1)(zx^2 + 3zx - x + 1). \quad (95)$$

For $0 < z < 1/9$, the four branch points in increasing order are

$$1, \quad r_-(z), \quad r_+(z), \quad 1/z, \quad r_{\pm}(z) = \frac{1-3z \pm \sqrt{(1-z)(1-9z)}}{2z}.$$

The middle pair collides at $(z, x, y) = (1/9, 3, 5/3)$, and there $\text{Disc}_y K = (x-1)(x-3)^2(x-9)/81$. This exact collision locates the critical parameters used next. The exponent itself will come from the covariance of the actual constrained walk, not from fitting the kernel.

[write] 5 October 2026. This subsection is not in Part I. Its collision point $(z, x, y) = (1/9, 3, 5/3)$ is Part I's Perron tilt (Section 3.1: $x = 3$, $y = 5/3$, eigenvalue $9 = 1/z$). Report 123 cites Britt and Beaton's equations (2.44)–(2.46); whether their (2.45)–(2.46) already print (92)–(94) was not checked, so these identities are presented as derived here (Section 27, item 1). Report 123's finite checker verifies both cleared functional equations and the eliminated equation at every coefficient through z^{65} , and the factorization (95) (123-logexp-checks-README.md); this is a check of the identities, which are proved by the geometric sums above.

17 A positive change of weights and its covariance

17.1 The Perron tilt and endpoint identity

[write] **5 October 2026.** Report 123's Section 3.1 is word for word Part I's Section 3.1: the step matrix (15), the tilt (16), the explicit laws (17), the endpoint identity (18) and the weight sum (19); it is printed there.

17.2 The color corrector

[write] **5 October 2026.** Report 123's Section 3.2 is Part I's Section 3.2 (the colour matrix (20), the corrector (21), the covariance (22) and the Perron-polynomial derivation), word for word except one clause. Report 123 writes "Its increment given a transition $c \rightarrow d$ is $v + h_d - h_c$, of conditional mean zero"; Report 125 writes that its mean conditional on the initial colour c is zero. Part I prints Report 125's wording, which names the right conditioning: the martingale property, and the covariances Γ_c , use the mean given the current colour. Given the transition $c \rightarrow d$ itself the mean is not zero in general: given $P \rightarrow P$ the increment is A_d with d geometric, $\mathbb{P}(d = k) = \frac{4}{9}(\frac{5}{9})^k$, so its mean is $(-\frac{1}{4}, \frac{5}{4})$. Report 123 uses only the first reading, so nothing in its proof changes.

17.3 Whitening and time reversal

Choose a real invertible W with $W\Sigma W^\top = I$. The image of the quadrant is a wedge of opening angle

$$\cos \theta = \frac{-\Sigma_{12}}{\sqrt{\Sigma_{11}\Sigma_{22}}} = \frac{2}{\sqrt{7}}. \quad (96)$$

Indeed the scalar product of the two transformed boundary vectors is computed with $W^\top W = \Sigma^{-1}$. We measure radii below by $\|s\|_W = |Ws|$ and use angular coordinate $0 \leq \phi \leq \theta$. The wedge's positive homogeneous Brownian harmonic function is $r^p \sin(p\phi)$, where $p = \pi/\theta$.

The reversed free process has probabilities

$$q_{dc}^*(-v) = \frac{\pi_c}{\pi_d} q_{cd}(v). \quad (97)$$

Stationarity proves its row sums are one. It has a primitive color matrix, exponential tails, zero stationary drift, and covariance Σ : a stationary reversed m -step displacement is the negative of a forward one, so its covariance growth rate is identical. Reversal of each retained path proves

$$\pi_c K_m((s, c), (t, d)) = \pi_d K_m^*((t, d), (s, c)). \quad (98)$$

The ratios of these two stationary masses are bounded between 1/2 and 2. The auxiliary killed kernels may be defined on the whole quadrant with both colors. Forward paths from o never enter a Q state with first coordinate zero. A dual path reaching such a state cannot subsequently reach the interior without leaving the quadrant, since every dual step out of Q decreases the first coordinate by one. Thus the successful dual arms and interior bridges below cannot introduce paths absent from the original tree.

[write] **5 October 2026.** Part I's Section 3.3 covers the same ground with the explicit whitening A of (3); (97) and (98) are (23) and (24), and the angle (96) is the one checked there with A .

18 Free functional and local limits

18.1 Checking the martingale functional limit hypotheses

For $R \rightarrow \infty$, let $Z^{(R)}(t) = R^{-1}Z_{\lfloor R^2 t \rfloor}$. Its predictable bracket is

$$\langle Z_i^{(R)}, Z_j^{(R)} \rangle(t) = R^{-2} \sum_{k < \lfloor R^2 t \rfloor} (\Gamma_{C_k})_{ij}.$$

The finite-state ergodic theorem makes this converge to $t\Sigma_{ij}$ uniformly on compact time intervals in probability, from either color. Every bracket jump is $O(R^{-2})$, with a fixed constant. Uniform exponential increment tails imply, for each fixed T ,

$$\mathbb{E} \max_{k \leq R^2 T} |Z_k - Z_{k-1}|^2 = O((\log R)^2), \quad \mathbb{E} \sup_{t \leq T} |\Delta Z^{(R)}(t)|^2 \rightarrow 0.$$

For completeness, integrate the union bound $\mathbb{P}(\max_{k \leq m} |\Delta Z_k| > u) \leq \min(1, Cme^{-cu})$ against $2u du$, splitting at $u = c^{-1} \log(Cm)$. Local square integrability holds by the same tails. These are exactly the predictable-bracket and maximal-jump requirements of Whitt [5, Theorem 2.1(ii)]. Thus $Z^{(R)}$ converges in the Skorohod topology to covariance- Σ Brownian motion. The corrector is bounded, so it vanishes on division by R . Polygonal interpolation gives the same limit, now in the uniform continuous-path topology on compact intervals. The dual satisfies the same conclusions. Explicitly its conditional drifts are $m_P^* = (1/2, -5/4)$ and $m_Q^* = (-1, 5/2)$, and a mean-zero corrector is $h_P^* = (2/3, -5/3)$, $h_Q^* = (-4/3, 10/3)$. It solves the same finite-state Poisson equation; its covariance is Σ by reversal.

Translation invariance and the finite number of colors make this convergence uniform for starting scaled positions in compact sets: otherwise a failing sequence has a convergent subsequence of positions and a constant subsequence of colors, contradicting the preceding limit. This compact-subsequence argument will also be used for the annuli.

18.2 Fourier inversion and removal of lattice periods

[write] **5 October 2026.** Report 123's Section 4.2 is word for word Part I's Section 9.1, with the local limit theorem (68); it is printed there.

18.3 An interior bridge bound

Lemma 18.1. *Let A, B be compact subsets of the open quadrant and $0 < \alpha < \beta < \infty$. There are $c > 0$ and n_0 such that*

$$K_m((x, c_0), (y, d_0)) \geq \frac{c}{n} \tag{99}$$

whenever $n \geq n_0$, $\alpha n \leq m \leq \beta n$, $x/\sqrt{n} \in A$, $y/\sqrt{n} \in B$, and either color is prescribed.

Proof. The free endpoint probability is at least c_1/n by (68). For a fixed $s < 1$, conditioning a free path on its final endpoint and color changes its density up to time $\lfloor sm \rfloor$ by the remaining endpoint kernel divided by that total probability. The numerator is at most $C/((1-s)m)$ and the denominator at least c_1/m . On compact scaled intermediate positions their ratio converges by (68)

to the Gaussian bridge density. The functional limit, this uniform bounded-density bound, and tightness of the free paths prove bridge convergence on $[0, s]$.

Applying the same argument to the reversed bridge controls the other end. Tightness on $[0, 2/3]$ and on $[1/3, 1]$ yields full-interval tightness, and the finite-dimensional limits identify the Brownian bridge. This argument is uniform in the stated compact sets and time ratios by subsequences. The segments joining A to B have uniformly positive distance from both coordinate axes, since the quadrant is convex. A fixed sufficiently thin tube about each segment therefore lies in the open quadrant. A Brownian bridge has a positive probability of lying in that tube, uniformly for the compact ranges of endpoints and durations; its centered fluctuation is a nonsingular Gaussian bridge and the duration range is compact. Portmanteau for the open tube gives a fixed positive lower bound for the conditioned discrete paths. Multiply it by c_1/n to obtain (99). $\square$

Only a free local limit and an interior bridge estimate have been proved. There is no assumed killed-cone local limit at the vertex or the boundary.

[write] **5 October 2026.** Lemma 18.1 is the model-specific form of Lemma 24.2 of `a377922-corner-polyhedra-schnyder` (`cps:log:lem:bridge`), with the same proof. Part I does not use it: its lower bounds come from the killed local limit (70).

19 Uniform annulus estimates

19.1 Brownian crossings and the time cutoff

Fix a closed angular interval $J \subset (0, \theta)$ with nonempty interior, containing the direction of $W(1, 1)$ in its interior. Fix a small $\delta > 0$, and consider starts with radii in $[1, 1 + \delta]$ in the whitened wedge. The harmonic measure of the radius- q arc, before a side, with bounded boundary values f , is

$$H_f(r, \phi) = \sum_{k \geq 1} b_k (r/q)^{kp} \sin(kp\phi), \quad b_k = \frac{2}{\theta} \int_0^\theta f(\psi) \sin(kp\psi) d\psi. \quad (100)$$

This follows by separating variables in the polar Laplace equation; boundedness at the vertex excludes negative radial powers. For $f = 1$, $b_1 = 4/\pi$. For f the indicator of a nonempty closed subarc of the interior of J , $b_1 > 0$. Since $|b_k| \leq 2\|f\|_\infty$, the terms $k \geq 2$ are uniformly $O(q^{-2p})$ on the stated starting radii. The first sine is uniformly positive for starting angles in J . There are therefore $c_B, C_B > 0$, independent of large q , such that

$$\begin{aligned} \mathbb{P}(\text{reach radius } q \text{ before a side}) &\leq C_B q^{-p}, & \text{all starting angles,} \\ \mathbb{P}(\text{exit on the chosen subarc of } J) &\geq c_B q^{-p}, & \text{starting angles in } J. \end{aligned} \quad (101)$$

For a start on a side the upper probability is zero: a nondegenerate Brownian normal coordinate immediately visits the exterior.

Uniform disk confinement has an exponential time tail. For example, from any position of modulus at most q , an increment over q^2 time units of modulus exceeding $3q$ forces departure from that disk and has a fixed positive probability a_B . The Markov property gives a bound $(1 - a_B)^{\lfloor t/q^2 \rfloor}$. Choose $T(q) < \infty$ large enough that this is smaller than a fixed multiple of q^{-p} at $t = T(q)q^2$. Thus the lower event in (101) can be required to finish before this time and retain, for example, half its lower bound. The upper bound can also be split at this time with an arbitrarily small multiple of q^{-p} lost in its tail.

19.2 Transfer to both discrete processes

Lemma 19.1. *For every $\eta > 0$, one can choose $q > 1$, $R_0 < \infty$, and $T < \infty$ such that the following hold for both the forward and dual killed walks. From any color and any starting position with $R \leq \|s\|_W \leq (1 + \delta)R$, $R \geq R_0$, the probability of reaching radius qR before killing is at most $q^{-p+\eta}$. If its angle is in J , the probability of reaching that radius before killing, in time at most $T(qR)^2$, with terminal angle in J and terminal radius at most $(1 + \delta)qR$, is at least $q^{-p-\eta}$.*

Proof. Fix q first. The functional limit transfers crossing probabilities on a fixed scaled time interval. For the upper estimate, the event that a continuous path has some witness time at which it reaches radius q and remains in the closed wedge up to that time is closed: along a convergent sequence of paths extract a convergent subsequence of witness times. Polygonal interpolations of retained discrete paths stay in the wedge by convexity. Portmanteau therefore gives the upper estimate on the truncated time interval, even along starting positions tending to a side.

A corresponding uniform discrete disk-confinement estimate follows from the free functional limit. For all sufficiently large r and all colors, a free displacement larger than $3r$ in $\lceil r^2 \rceil$ steps has probability at least some $a > 0$ independent of r . From any point in the radius- r disk it forces exit. Hence

$$\mathbb{P}(\text{stay in the radius-}r \text{ disk for } m \text{ steps}) \leq (1 - a)^{\lfloor m/\lceil r^2 \rceil \rfloor}. \quad (102)$$

This bounds late crossings after the truncated interval and removes the time cutoff in the upper bound.

For the lower estimate choose the smaller subarc used above and a strict time bound. A Brownian path succeeding before a side, starting inside, has positive distance from the sides up to its finite exit time; it also crosses the outer circle immediately almost surely. Boundary arc endpoints have zero exit mass. One may exhaust the successful event by open tubes that cross the circle and end in the interior of J . Portmanteau then gives its lower bound. Exponential tails give

$$\mathbb{P}\left(\max_{j \leq CR^2} |v_j| > \varepsilon R\right) \leq CR^2 A_0 e^{-b'_0 \varepsilon R} \longrightarrow 0$$

for fixed $C, \varepsilon > 0$, so the relative overshoot can also be bounded by δ . This applies to both processes. Compactness of the scaled starting annulus, a subsequence argument, and the finite colors make the bounds uniform. After the time-tail choices the transferred bounds are $C_1 q^{-p}$ above and $c_1 q^{-p}$ below, with positive fixed c_1, C_1 independent of large q . Choose q so that $C_1 \leq q^\eta$ and $c_1 \geq q^{-\eta}$; then choose R_0 large enough for that fixed q . The resulting time constant T and scale threshold may depend on η, J, δ and the two chains, never on later n . $\square$

[write] **5 October 2026.** Section 19 is the model-specific form of the crossing bounds of Theorem 24.1's proof in `a377922-corner-polyhedra-schnyder` (its Section 24.4, `cps:log:sec:crossings`); the two differ only in how the overshoot at each radius is controlled (there by the jump bound $\sqrt{R}$, here by the relative bound δ).

20 The logarithmic upper bound

20.1 Survival from slowly growing starting sets

Lemma 20.1. *For each $L < \infty$ and $\varepsilon > 0$, for either chain,*

$$\sup_{\|s\|_W \leq L \log n, c} \mathbb{P}_{s,c}(\tau > n) \leq C_{L,\varepsilon} n^{-p/2+\varepsilon} \quad (103)$$

for all sufficiently large n , where τ is the killing time.

Proof. Choose a target error power $K > p/2 + 1$. For $B_n = A \log n$, the uniform exponential tails and the union bound show that the event

$$G_n = \{\|v_j\|_W \leq B_n \text{ for } 1 \leq j \leq n\}$$

has complement at most Cn^{-K} if A is sufficiently large. This cutoff is imposed only on the first n steps of the original process; we do not define a truncated process or change its covariance.

Start annular comparison at $r_0 = (\log n)^2$ and radii $r_j = q^j r_0$, where q, R_0 come from Lemma 19.1 with small $\eta > 0$. The initial position lies inside r_0 for large n . On G_n , the first hit of any r_j by time n overshoots it by at most $B_n = o(r_j)$; for large n it cannot skip the next scale. After each such first hit, the next crossing probability is at most $q^{-p+\eta}$. In applying the Markov property we drop any restrictions on future jumps, which only increases probability. Therefore, if $R \geq qr_0$,

$$\mathbb{P}(\text{reach } R \text{ before killing and by time } n, G_n) \leq C_q (R/r_0)^{-p+\eta}. \quad (104)$$

Taking the last scale below R changes only C_q .

A path surviving to time n either reaches R by then or stays in the radius- R disk. Put $R = \sqrt{n/(M \log n)}$. Equation (102) is $O(n^{-K})$ when M is sufficiently large (depending on K). Combining with (104) yields

$$\mathbb{P}(\tau > n) \leq Cn^{-(p-\eta)/2} (\log n)^{(5/2)(p-\eta)} + Cn^{-K}.$$

The power of $\log n$ is immaterial; it is displayed to make the scale loss explicit. Choose η with $\eta/2 < \varepsilon$ and absorb this fixed logarithmic power into the remaining positive power of n . All choices are uniform in the specified initial set, proving the lemma. $\square$

20.2 A reverse endpoint bound and summing the weights

Split $n = n_1 + n_2 + n_3$ into three integers comparable to n . For an endpoint (t, d) with $\|t\|_W \leq L \log n$, use the Markov property, remove killing in the middle, and apply (68):

$$\begin{aligned} K_n(o, (t, d)) &\leq \frac{C}{n} \sum_{x,c} K_{n_1}(o, (x, c)) \sum_{y,e} K_{n_3}((y, e), (t, d)) \\ &\leq \frac{2C}{n} \mathbb{P}_o(\tau > n_1) \mathbb{P}_{t,d}^*(\tau > n_3). \end{aligned}$$

The second line is exactly the reversal identity (98), with $\pi_d/\pi_e \leq 2$. Applying Lemma 20.1, with half of the desired error on each side, gives for every $\varepsilon > 0$

$$K_n(o, (t, d)) \leq C_{L,\varepsilon} n^{-p-1+\varepsilon}. \quad (105)$$

The norm $\|t\|_W$ is equivalent to $a + b$ on the quadrant, so

$$\sum_{\|t\|_W > L \log n, d} g(t, d) \leq C e^{-cL \log n}.$$

Choose L large enough that this is at most Cn^{-p-2} . Use $K_n \leq 1$ on this tail, and (105) and (19) on its complement. The exact identity (18), with $n - 1$ in place of n , proves the upper half of (89).

[write] **5 October 2026.** Lemma 20.1 is uniform over starts at radius at most $L \log n$. The statement of Theorem 24.1 of `a377922-corner-polyhedra-schnyder` gives survival from a fixed start only, but its proof gives this uniformity (Proposition 26.2(i)). Part I's global bound (67) is sharper (a power of $1 + |s|$ in place of n^ε) and covers every start.

21 The lower bound at the original integer time

The fixed endpoint used here is $e = (1, 0, P)$ in shifted coordinates, and its weight in (18) is $g(e) = 1/6 > 0$.

21.1 Two positive probability seed paths

The forward chain from o can reach (H, H, P) using H steps $A_0 = (1, 0)$ and H steps $A_1 = (0, 1)$ in any order. These stay in the quadrant and each has positive probability. For the dual from e the following explicit path is valid:

$$(1, 0, P) \rightarrow (1, 0, Q) \rightarrow (0, 2, P) \rightarrow (2, 0, Q) \rightarrow (1, 3, P). \quad (106)$$

The four steps reverse, respectively, C_1, D_2, C_3, D_3 . From a state of color P , the dual pair $(2, -2) : P \rightarrow Q$ and $(-1, 3) : Q \rightarrow P$ has net displacement $(1, 1)$. Starting at $(1, 3, P)$ all its intermediate states remain in the quadrant. Thus the dual too reaches arbitrarily large radii with direction tending to that of $W(1, 1)$, by a fixed positive-probability path. Choose the two seed paths once so that their endpoints have radius at least R_0 and angle in the interior of J . Their possibly different radii and their lengths are constants independent of n .

21.2 Stopping arms and exact length gluing

Fix small $\eta > 0$ and the constants of Lemma 19.1. Starting at each seed endpoint, impose successive successful lower annulus events at scales multiplied by q . Stop at the first hit of the largest chosen radius at most $d\sqrt{n}$. Its value lies between $d\sqrt{n}/q$ and $d\sqrt{n}$ for large n . Every final endpoint has angle in J and relative overshoot at most δ . On division by $\sqrt{n}$, the forward and dual endpoints therefore belong to fixed compact subsets A, B of the open quadrant.

The number of crossings in each arm is $\frac{1}{2} \log n / \log q + O(1)$, so the mass of each successful arm is at least $c_\eta n^{-(p+\eta)/2}$. Its total allowed time is at most

$$\text{seed length} + T \sum_{j \leq k} r_j^2 \leq \text{seed length} + \frac{T}{1 - q^{-2}} d^2 n.$$

Choose $d > 0$ small enough that the last coefficient is less than $1/8$. For sufficiently large n , each arm then lasts at most $n/4$ including its seed. The final time is its first-hit stopping time of the

chosen final radius: no earlier point of an earlier annulus can exceed that radius on a successful arm. If needed increase q so that $1 + \delta < q$.

For every successful forward arm with time $t \leq n/4$ and terminal state (x, c) , and every successful dual arm with time $u \leq n/4$ and terminal state (y, d) , insert a forward killed bridge of exactly $m = n - t - u$ steps from (x, c) to (y, d) . Since $n/2 \leq m \leq n$, Lemma 18.1 gives mass at least c/n uniformly. Reverse the dual arm to obtain a suffix ending at e . By (98), its weight differs from the dual-arm weight by π_P/π_d , a factor between 1 and 2.

There is no overcounting in this gluing. From a resulting length- n path, the forward arm is recovered as the first forward crossing of its fixed target radius, and the suffix as the first backward crossing of the dual target radius, with backward traversal starting at e . All these radii remain centered at the coordinate origin. They are separated because both stopping lengths are at most $n/4$. The prescribed seed and intermediate crossing conditions restrict these recovered arms; they do not create new decompositions. Summing over their actual integer stopping times and endpoints therefore gives

$$K_n(o, e) \geq c_\eta n^{-p-1-\eta}. \quad (107)$$

No renewal-time replacement has occurred. This is a bound at every sufficiently large original integer n . Insert the positive fixed weight $g(e)$ in (18) and reindex $n - 1$ to obtain the lower half of (89). Together with the upper bound it proves the logarithmic coefficient law of Theorem 15.1.

[write] **5 October 2026.** The fixed endpoint $e = (1, 0, P)$ is not at the origin. Theorem 24.1 of `a377922-corner-polyhedra-schnyder` is stated for endpoints at the origin, where this walk has no dual seed and $K_n(o, ((0, 0), b)) = 0$ for $n \geq 1$ (Remark 26.1); this section proves, for this walk, the statement of Proposition 26.2(ii) at (o, e) .

22 The unique singularity on the critical circle

[write] **5 October 2026.** Report 123’s Section 8 (“Uniform return loops” and “The operator resolvent away from the positive point”) is Part I’s Section 12 without its opening corollary: the return loops (81), the resolvent (82) and the representation (83). It is printed there. Where Part I cites (9) for the radius $1/9$, Report 123 cites (89); Cauchy–Hadamard needs only that.

23 Irrationality and the arithmetic obstruction

[write] **5 October 2026.** Report 123’s Section 9 (“The irrational exponent”, “A power bound for the fifth derivative”, “The precise G function implication”, “Descent from complex to rational coefficients”) is Part I’s Section 13, word for word except that its Section 9.2 bounds the summands by (89) where Part I uses (9); the bounds $n^{5-\kappa \pm \varepsilon} t^n$ that the argument needs follow from either. It is printed there, with (84), (85) and (86); its conclusion is this Part’s (90).

24 Inversion and the remaining open leading law

Appending the new maximum n to any length- n avoider is legal and injective: a newly formed triple would end with a value larger than all earlier entries and cannot satisfy (87). Thus a_n is nondecreasing. It is unbounded by (89), so $N(y)$ is well-defined for large y .

Write $\lambda = \log 9$, $L = \log y$, and $m = N(y)$. The logarithmic law is exactly

$$\log a_n = \lambda n - \kappa \log n + o(\log n).$$

Since $a_{m-1} < y \leq a_m$, first division by m gives $L/m \rightarrow \lambda$. Use this in both inequalities of the consecutive-integer sandwich to obtain

$$L = \lambda m - \kappa \log m + o(\log m), \quad \log m = \log L - \log \lambda + o(1).$$

The constant $-\kappa \log \lambda$ and the one-step displacement are $o(\log L)$, yielding (91). This is an integer threshold statement with a genuinely unbounded allowed remainder. No bounded inverse constant, ceiling prescription with bounded error, differentiable remainder, or all-order inverse follows from the proved law.

A full leading equivalent would require a further result: for example, a finite-state Markov-additive killed-cone endpoint local limit, with uniform control strong enough to sum (18). An iid cone theorem cannot be substituted, since the color-conditioned drift is not zero and the position process alone is not the required iid walk. The present annulus method avoids that missing stronger result and loses arbitrarily small powers at each scale. It proves exactly the logarithmic exponent and the stated consequences. In particular it leaves bounded log-periodic amplitudes and logarithmic corrections unresolved.

[write] **5 October 2026.** The last paragraph describes Report 123 alone, and Part I supplies the further result it asks for, for this walk: a killed local limit at fixed endpoints, (7), with the uniform endpoint bound (8) that sums (18). It gives the equivalent (9), so bounded log-periodic amplitudes and logarithmic corrections are excluded; the inverse has error $O(1)$ (Corollary 1.2). Part I also avoids an iid cone theorem (its Section 1, “Scope and attribution”). A ceiling rule with bounded error is proved only as the bracket (12); an exact rule and an all-order inverse remain open (Section 14.1, item 4).

25 Reproducibility and limits of finite checks

The companion package supplies this editable source, the rendered PDF, exact integer data, and offline finite checks. It separates two forms of evidence. Rational algebra and finite enumeration can be replayed exactly; the uniform limits, annulus argument, analytic continuation, and G-function theorem application are mathematical proofs in the text, not consequences of matching a finite coefficient list.

The independent checker documents its precise finite ranges in `checks/README.md`. Its active guards check the generating rules, critical algebra, color corrector, effective covariance, return loops, and declared fixtures. It is required to retain those checks under optimized Python. Negative tests deliberately mutate mathematical data and the closed inventories. The complete package has a bitwise inventory and rejects extra files, missing files, changed bytes, and symbolic links. These inventories are integrity records, not cryptographic signatures.

The build creates the PDF twice in clean external directories, rejects overfull boxes and unresolved references, and requires byte equality. The full replay repeats finite checks in normal and optimized Python, rebuilds the PDF, creates a deterministic ZIP, extracts it into a fresh directory, reruns checks and rebuilds, and requires the extracted tree to repack identically. Build details and commands are in `README.md`. No downloaded source papers, numerical fit files, intermediate render images, or private working material form part of the reader package.

[write] **5 October 2026.** In this repository Report 123's `checks/README.md` is `123-logexp-checks-README.md` and its checker `code/123-logexp-checks-verify.py`; the PDF and the inventories are not shipped, and its `README.md` and `report123.tex` are not shipped either (this Part prints the latter). Report 125's package contains these checks as `checks/legacy/` and reruns them; they are shipped once. The report's README gives rerun routes on a copy of the delivery.

26 Relation to the cone theorem of a377922

[write] **5 October 2026.** This section is added in the write. Theorem 24.1 of `a377922-corner-polyhedra-schnyder` (bundle Report 112, `cps:log:thm:cone`) states this Part's method for a translation-invariant Markov-additive chain on $\mathbb{Z}^2 \times E$, E finite, killed on leaving the closed quadrant, under five hypotheses: (H1) a primitive phase matrix with positive stationary law; (H2) a uniform exponential moment of the steps; (H3) zero stationary drift and a positive definite effective covariance Σ ; (H4) no peripheral Fourier eigenvalue except the simple Perron eigenvalue at 0; and (H5), for endpoint phases a, b at the spatial origin, a compact interval $J \subset (0, \theta)$ of positive length such that for every $R > 0$ a finite positive-probability legal path from $(0, a)$ reaches a point x with $|Wx| \geq R$ and $\arg(Wx) \in J$, and the same from $(0, b)$ for the stationary dual. Under (H1)–(H5) it proves $K_n((0, a), (0, b)) = n^{-1-\nu+o(1)}$ with $\nu = \pi/\theta$, and under (H1)–(H4) the survival bound $\mathbb{P}_{(0,c)}(\tau > n) \leq C_\epsilon n^{-\nu/2+\epsilon}$. Here $E = \{P, Q\}$, the chain is the tilted walk (17), its dual is (23), which is that theorem's dual, and $\nu = p$. “Legal” means that every visited lattice point lies in the closed quadrant. Arguments of whitened vectors are measured from the side We_1 of the whitened wedge, which is then $\{0 < \arg < \theta\}$; this is the convention under which (H5) there writes $J \subset (0, \theta)$.

Remark 26.1 ([write], 5 October 2026: the hypotheses for this walk). (a) The tilted walk of Part I, Section 3.1, satisfies (H1)–(H4), and its wedge angle in the sense of Theorem 24.1 is $\theta = \arccos(2/\sqrt{7})$.

(b) The forward half of (H5) holds at the phase P of the origin, but the dual half fails at both phases $b \in \{P, Q\}$ of the origin: from $((0, 0), Q)$ the dual has no legal step, and from $((0, 0), P)$ its only legal step goes to $((0, 0), Q)$.

(c) For every state (x, a) and every $n \geq 2$, $K_n((x, a), ((0, 0), b)) = 0$ for both b ; for $n = 1$ it is positive only for $(x, a) = ((0, 0), Q)$ and $b = P$. In particular $K_n(o, ((0, 0), b)) = 0$ for all $n \geq 1$.

So Theorem 24.1 makes no claim about this walk at any pair of endpoints its statement allows, and its first display would be false here if (H5) were dropped; only its survival bound applies, from o .

Proof. (a) (H1): the colour matrix P_0 of (20) has positive entries and $\pi = (2/3, 1/3)$. (H2): by (17) the step lengths have geometric tails with ratios $5/9$ and $3/5$ from either colour. (H3): $\pi_P m_P + \pi_Q m_Q = 0$ (Section 3.2), and the effective covariance in the sense of Theorem 24.1, $\sum_c \pi_c \Gamma_c$ for the corrected martingale increments, is Σ of (22), with $\Sigma_{11} = 7/3 > 0$ and $\det \Sigma = 25/3 > 0$. (H4): Section 9.1 shows that $L(u)$ has a unit-modulus eigenvalue only at $u \equiv 0$; since its spectral radius is at most one, it is less than one at every nonzero u , and $L(0) = P_0$ has the eigenvalues 1 and $1/4$. Finally $\rho = \Sigma_{12}/\sqrt{\Sigma_{11}\Sigma_{22}} = (-10/3)/(5\sqrt{7}/3) = -2/\sqrt{7}$, so $\arccos(-\rho) = \arccos(2/\sqrt{7})$.

(b) The forward seeds of Section 7.4 start at $o = ((0, 0), P)$. For the dual, (23) gives a step of positive probability from (s, d) with displacement $-v$ to colour c exactly when the forward walk has an edge $c \rightarrow d$ with displacement v . By (14) the edges into Q are the D_j , $v = (1, -j)$, from either colour, so every dual step out of a Q state has displacement $(-1, j)$. The edges into P are the A_d from P , $v = (1 - d, d)$ with $d \geq 0$, and the C_d from Q , $v = (1 - d, d - 1)$ with $d \geq 1$; the dual steps out of a P state therefore have displacement $(d - 1, -d)$ (to P) or $(d - 1, 1 - d)$ (to Q). From the origin, $(-1, j)$ leaves the quadrant; $(d - 1, -d)$ would need $d \geq 1$ and $d \leq 0$; and $(d - 1, 1 - d)$ stays

in the quadrant only for $d = 1$, where it is the zero step to $((0, 0), Q)$. Hence every legal dual path from $((0, 0), b)$ stays at the origin and has length at most one, and no such path reaches $|Wx| \geq R$ for any $R > 0$.

(c) By the reversal identity (24), $\pi_a K_n((x, a), ((0, 0), b)) = \pi_b K_n^*((0, 0), b), (x, a))$. By (b), $K_n^*((0, 0), b), \cdot) = 0$ for $n \geq 2$, and $K_1^*((0, 0), b), \cdot)$ is positive only for $b = P$, at $((0, 0), Q)$. Since o has colour P , $K_n(o, ((0, 0), b)) = 0$ for every $n \geq 1$. $\square$

This agrees with Part I: the dual harmonic function vanishes at the origin, $V^*((0, 0), b) = 0$, since the dual from there dies within two steps; Part I's Section 3.3 already notes that the auxiliary Q states with first coordinate zero, $((0, 0), Q)$ among them, may have zero harmonic value.

Proposition 26.2 ([write], 5 October 2026: Theorem 24.1 at fixed endpoint states). *Let (S_n, C_n) be a chain as in Theorem 24.1 of `a377922-corner-polyhedra-schnyder` that satisfies (H1)–(H4), with its stationary dual, killed kernels K_n, K_n^* , killing times τ, τ^* , whitening $W = \Sigma^{-1/2}$, wedge angle θ and $\nu = \pi/\theta$.*

(i) *For every $\epsilon > 0$ and $D' < \infty$ there is C such that, for all $n \geq 1$, all phases a, b and all $x, y \in \mathbb{Z}_{\geq 0}^2$ with $|x|, |y| \leq D' \log(n + 2)$,*

$$\mathbb{P}_{(x,a)}(\tau > n) + \mathbb{P}_{(y,b)}^*(\tau^* > n) \leq Cn^{-\nu/2+\epsilon}, \quad K_n((x, a), (y, b)) \leq Cn^{-1-\nu+\epsilon}. \quad (108)$$

(ii) *Fix states (x, a) and (y, b) with $x, y \in \mathbb{Z}_{\geq 0}^2$. Suppose that there is a compact interval $J \subset (0, \theta)$ of positive length such that, for every $R > 0$, a finite positive-probability legal path of the chain from (x, a) , and one of the dual from (y, b) , reach points z with $|Wz| \geq R$ and $\arg(Wz) \in J$. Then for every $\eta > 0$ there is $c_\eta > 0$ with*

$$K_n((x, a), (y, b)) \geq c_\eta n^{-1-\nu-\eta} \quad \text{for all sufficiently large } n, \quad (109)$$

and so $K_n((x, a), (y, b)) = n^{-1-\nu+o(1)}$. For $x = y = 0$ this is Theorem 24.1.

Proof. We follow the proof of Theorem 24.1 in `a377922-corner-polyhedra-schnyder` (its Sections 24.5 and 24.6, `cps:log:sec:upper` and `cps:log:sec:lower`, in its notation: $X_n = WS_n$, the radii r_j , the first hitting times σ_j , the events A_j) and list every place where it uses the endpoints.

(i) *Survival.* Fix η , then $q, T(q)$ and $R_0(q)$ as in its display (24.14) (`cps:log:eq:chooseq`), then $M > \nu + 2$, $D, L_n = D \log(n + 2)$, $r_0(n) = \max\{R_0, L_n^2\}$ and $r_j = q^j r_0(n)$, as there. The start enters that argument only through the first radius: if $|X_0| < r_0(n)$, then σ_0 is the first hit of radius $r_0(n)$ from inside, $\mathbb{P}(A_0) \leq 1$, and the overshoot bound $L_n \leq \sqrt{r_j}$ on A_j , the inclusions $A_{j+1} \subset A_j$ and the strong Markov steps behind (24.16) (`cps:log:eq:annularupper`) do not involve X_0 . The final dichotomy (a bad jump, confinement in the disk of radius $R_n = \sqrt{n/(B \log(n + 2))}$, or the event A_J with $r_J \leq R_n < r_{J+1}$) holds for every start in that disk, and the confinement bound (24.9) (`cps:log:eq:confine`) holds from every point of the disk. Hence (24.17) (`cps:log:eq:survival-proof`) holds uniformly over all starts with $|X_0| < r_0(n)$ and both phases, with constants that depend on η and the chain only. If $|x| \leq D' \log(n + 2)$ then $|Wx| \leq \|W\| D' \log(n + 2) < D^2 \log^2(n + 2) \leq r_0(n)$ for all n beyond a threshold that depends on D', D and W ; smaller n are absorbed into C . Choosing $\eta < \epsilon$ gives the survival half of (108). The dual satisfies (H1)–(H4) (it preserves centring, exponential tails and the effective covariance, and its Fourier matrix is $\text{diag}(\pi)^{-1} F(-t)^\top \text{diag}(\pi)$, as shown there), so the same holds for it.

Kernel. Write $n = n_1 + n_2 + n_3$ with $n_1, n_2, n_3 \geq \lfloor n/3 \rfloor$. By Chapman–Kolmogorov, and since the killed middle kernel is at most the unrestricted one, which is at most C/n_2 by the local limit theorem (24.5) (`cps:log:eq:llt`),

$$K_n((x, a), (y, b)) \leq \frac{C'}{n} \sum_{u, c} K_{n_1}((x, a), (u, c)) \sum_{w, d} K_{n_3}((w, d), (y, b)).$$

The first sum is $\mathbb{P}_{(x, a)}(\tau > n_1)$. By the reversal identity (24.6) (`cps:log:eq:reversal`) the second is $\pi_b \sum_{w, d} K_{n_3}^*((y, b), (w, d))/\pi_d \leq (\pi_b / \min_d \pi_d) \mathbb{P}_{(y, b)}^*(\tau^* > n_3)$. For $n \geq 6$ one has $\log(n+2) \leq 2\log(n_i+2)$, so $|x|, |y| \leq 2D' \log(n_i+2)$, and the survival half, with $2D'$ and $\epsilon/2$, bounds both probabilities by $Cn^{-\nu/2+\epsilon/2}$. This proves the kernel half.

(ii) *Lower bound.* The argument of its Section 24.6 (`cps:log:sec:lower`) uses the endpoints in four places only. (1) It fixes one legal seed for each chain with terminal radius at least R_0 and terminal angle in J ; the hypothesis of (ii) supplies them from (x, a) and, for the dual, from (y, b) . (2) For large n the final radius $r_J > \delta\sqrt{n}/q$ exceeds the largest radius visited by the fixed seed; the seed now starts at x (or y), but its largest radius is still a fixed finite number, so this holds for large n , and every accepted prefix ends at the first global hit of its final radius. (3) Reversing the accepted dual prefix gives a suffix ending at (y, b) , with weight changed by a ratio of stationary masses, bounded above and below. (4) A glued path determines its two prefixes as the first forward hit and the first backward hit of the two final radii, by (2) and because both prefixes have length at most $n/4$. The crossing estimates (24.12) (`cps:log:eq:cross-lower`) concern states at radius at least R_0 , and the bridge bound (Lemma 24.2 there) concerns states at scale $\sqrt{n}$ in a fixed compact subset of the open quadrant; neither involves the endpoints. Hence the escape-mass bound (24.19) (`cps:log:eq:escapemass`) holds for the prefixes from (x, a) and (y, b) , and the gluing gives $K_n((x, a), (y, b)) \geq (c/n)F_nG_n \geq c_\eta n^{-1-\nu-\eta}$, which is (109). Together with (i) at fixed x, y , this gives $K_n((x, a), (y, b)) = n^{-1-\nu+o(1)}$. $\square$

Corollary 26.3 (`[write]`, 5 October 2026: Report 123's exponent from Proposition 26.2). *For the tilted walk of Part I, Section 3.1, $K_n(o, e) = n^{-1-p+o(1)}$ with $e = ((1, 0), P)$, and Report 123's bounds (89) follow: for every $\epsilon > 0$, $c_\epsilon 9^n n^{-\kappa-\epsilon} \leq a_n \leq C_\epsilon 9^n n^{-\kappa+\epsilon}$ for large n .*

Proof. (H1)–(H4) hold by Remark 26.1(a), with $\nu = p$. Let $\phi_0 = \arg(W(1, 1))$; since $(1, 1)$ lies in the open quadrant, $\phi_0 \in (0, \theta)$, and we take $J = [\phi_0 - \delta', \phi_0 + \delta'] \subset (0, \theta)$ for a small $\delta' > 0$. From o , H steps A_0 followed by H steps A_1 reach $((H, H), P)$ through legal states with positive probability (Section 7.4), at whitened angle ϕ_0 and radius $H|W(1, 1)| \rightarrow \infty$. From e , the dual seed (57) reaches $((1, 3), P)$, and k repetitions of the dual pair $(2, -2) : P \rightarrow Q, (-1, 3) : Q \rightarrow P$ reach $((1, 3) + k(1, 1), P)$ through legal states (Section 7.4); the whitened angle tends to ϕ_0 and the radius to infinity as $k \rightarrow \infty$. So the hypothesis of Proposition 26.2(ii) holds at (o, e) , and $K_n(o, e) \geq c_\eta n^{-1-p-\eta}$. By (18) with $g(e) = 1/6$,

$$a_{n+1} = 2 \cdot 9^n \sum_{t, d} g(t, d) K_n(o, (t, d)) \geq \frac{1}{3} 9^n K_n(o, e) \geq \frac{c_\eta}{3} 9^n n^{-1-p-\eta}.$$

For the upper bound, fix D' with $D' \log(5/3) > p + 3$. Proposition 26.2(i) and (19) bound the endpoints with $|t| \leq D' \log(n+2)$ by $\frac{45}{8} Cn^{-1-p+\epsilon}$. For the others use $K_n \leq 1$, $g((a, b), d) \leq (3/5)^{a+b}/r_d$, $\sum_d 1/r_d = 3/2$ and $|t| \leq a + b$: their total weight is at most $\frac{3}{2} \sum_{m > D' \log(n+2)} (m+1)(3/5)^m = O(n^{-p-2})$. Hence $a_{n+1} \leq C_\epsilon 9^n n^{-1-p+\epsilon}$. Since $\kappa = 1 + p$ and $(n+1)/n \rightarrow 1$, reindexing gives (89). $\square$

This is the argument of Sections 20–21 with Lemma 20.1 and the gluing replaced by Proposition 26.2: Report 123’s exponent is an instance of the proof of Theorem 24.1, and with Proposition 26.2 it becomes a corollary of a stated theorem. Report 123’s non-D-finiteness and circle statements then follow as in Sections 12–13, and its non-D-finiteness argument is an instance of the criterion of Remark 27.1 there (`cps:log:rem:criterion`).

Remark 26.4 (**[write]**, 5 October 2026: what Part I adds to Questions Q1 and Q3 of a377922). Question Q1 there (`cps:log:q:amplitude`) asks whether, under (H1)–(H5), $K_n((0, a), (0, b)) \sim \kappa V_a(0)V_b^*(0) n^{-1-\nu}$ with positive phase-indexed killed harmonic functions, and names as missing a harmonic comparison without a cycle buffer, or a phase-sensitive killed local limit theory. For this one walk Part I proves such a statement at every fixed pair of states, (7), with positivity of V and V^* in place of (H5) ($V(o) > 0$ and $V^*(e) > 0$, Section 7.4; $V^*((0, 0), b) = 0$, consistent with Remark 26.1(c)), with no cycle buffer, by exactly such a local limit theory: deep entrance, an angle-uniform Brownian limit, last-window smoothing and midpoint reversal. It does not generalize as written, and Report 125 says that no general theorem is claimed (Section 14). Part I uses $p > 4$ to make the extended U of class C^3 at the origin (Section 7.2), $q = p - 3 \in (1, 2)$ in the potential bound (43), and $p > 2$ for the mean exit time (39). It is a third route to amplitudes beside Part II of a377922–corner–polyhedra–schnyder (comparison with an iid cone walk through a fixed cycle buffer) and that report’s Part III (annuli, logarithmic scale only).

Question Q3 there (`cps:log:q:seeds`) asks for the exponent, if any, at endpoints without seeds. This walk is an example: at both phases of the origin the dual has no seed (Remark 26.1(b)), and K_n into the origin vanishes for every $n \geq 2$ (Remark 26.1(c)), so there is no exponent. The question remains open in general.

27 Further questions and research

[write] 5 October 2026. This section is added in the write. Under Vladimir’s standing rule of 4 October 2026 it collects the claims of this Part that are stated without proof, with source, sketch and what is missing. Items 1–7 of Section 14.1 concern both Parts and are not repeated: digits of C_A , rates and corrections, a local expansion at $1/9$, the ceiling rule, effective onsets (including this Part’s n_ε and the n_0 of Lemma 18.1), a general amplitude theorem, and external inputs. The open items that Report 123 lists in Sections 15 and 24 (the equivalent and its constant, log-periodic modulation, an inverse with bounded error, the killed local limit that “a full leading equivalent would require”) are settled or re-scoped by Part I (dated notes there). The intake found no false claim in Report 123; the clause discussed in Section 17.2 is a matter of wording.

1. **The functional equations in Britt and Beaton.** Report 123 cites Britt and Beaton’s Section 2.5, equations (2.44)–(2.46), for its generating tree, and derives (92)–(95) itself. Do their (2.45)–(2.46) already print these identities, or equivalent ones? Missing: a comparison with arXiv:2512.21943v3, which was not retrieved at the write. Until then the identities are credited to Report 123 as derived there.
2. **An analytic route through the kernel.** Report 123 locates the critical point by the collision of the branch points $r_\pm(z)$ at $(1/9, 3, 5/3)$ but takes the exponent “from the covariance of the actual constrained walk, not from fitting the kernel”, and assumes no global contraction for the affine relation between kernel branches (Section 16.2). Can the kernel equation (94) give an independent proof of the exponent, the local expansion at $1/9$ of Section 14.1, item 3, or a formula for C_A ? This is a research question; neither Part claims anything about it.

3. **Endpoints without seeds.** Question Q3 of `a377922-corner-polyhedra-schnyder` remains open in general; for this walk it is answered by Remark 26.4: no exponent at the origin, because K_n vanishes there.
4. **Steps of this Part not re-checked at the write.** The bridge tightness in the proof of Lemma 18.1 and the uniform transfer from Brownian motion near the sides of the wedge in Lemma 19.1 are given with the arguments printed above; the intake read them as sound, as it read their general forms in `a377922-corner-polyhedra-schnyder` (its Question Q4). The locator of Whitt’s Theorem 2.1(ii) was not checked against the paper.

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